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Hallucination detection, verification, and correction in generative AI: A comprehensive survey

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ABSTRACT

The rapid adoption of large language models (LLMs) and generative artificial intelligence (AI) systems has brought AI hallucinations to the forefront of contemporary natural language processing research. AI hallucinations refer to statements made by the system which are either factually incorrect, illogical, or not substantiated by any credible sources but sound coherent and fluent on the face of it. These behaviors raise serious problems when using AI in high-stakes applications such as medicine, teaching, legal matters, finance, or science, where accurate facts are required. The present survey aims to provide an extensive overview of current techniques employed to detect, verify, mitigate, or correct hallucinations in generative AI models. Initially, we provide a taxonomy of different types of AI hallucinations. Subsequently, we examine major detection and verification paradigms, including retrieval-augmented generation (RAG), factuality evaluation metrics, automated fact-checking, natural language inference (NLI), knowledge-grounded reasoning, multi-model verification, and self-reflection-based approaches. The survey further reviews widely used benchmark datasets, evaluation frameworks, and emerging multi-layer verification architectures for improving AI trustworthiness. In addition, enabling technologies such as transformer-based language models, dense retrieval systems, knowledge graphs, credibility assessment mechanisms, and truth-scoring frameworks are analyzed from the perspective of reliable content verification. Lastly, we highlight important limitations of existing approaches along with potential future research directions, such as scaling verification, multilingual reasoning, real-time verification, and privacy-enhanced trusted artificial intelligence systems. In summary, this survey offers a comprehensive guide to researchers and developers aiming to design reliable, interpretable, and trustworthy generative AI models.

1. Introduction

The advent of large language models (LLMs) has profoundly transformed how humans acquire, analyze, and convey information. While AI text generation stems from rule based NLP systems, the transformer model by Vaswani et al. (2017) enabled modern systems like GPT 4 (Achiam et al., 2023) and LLaMA 2 (Touvron et al., 2023) to produce authoritative text indistinguishable from human writing. However, a severe failure mode is hallucination, where a system generates grammatically flawless text that is factually incorrect, logically inconsistent, or temporally implausible (Huang et al., 2025a; Tonmoy et al., 2024). Crucially, contemporary LLMs present hallucinations as confidently as verified facts (Anh-Hoang et al., 2025; Pesaranghader and Li, 2026). The consequences are severe: false clinical guidelines harm patient health, legal professionals cite non existent rulings, and erroneous data erodes scientific credibility. Ultimately, misinformation diminishes trust in science, journalism, and democratic institutions (Broda and Strömbäck, 2024; Tomassi et al., 2025), making AI reliability a critical social issue.

Extensive literature addresses verification aspects, including hallucination taxonomy, retrieval-augmented generation (RAG) (Lewis et al., 2020; Shuster et al., 2021), factuality metrics (Min et al., 2023b; Fabbri et al., 2022; Durmus et al., 2020), fact checking (Guo et al., 2022; Schlichtkrull et al., 2023), cross-model validation (Adlakha et al., 2024), and trustworthy AI (Amodei et al., 2016; Bommasani et al., 2021; Weidinger et al., 2021). However, these focus on narrow subsets. RAG reduces but cannot eliminate hallucination (Niu et al., 2024). Factuality

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metrics like FActScore (Min et al., 2023b), QAFactEval (Fabbri et al., 2022), and FEQA (Durmus et al., 2020) gauge consistency but lack full stack implementation. Furthermore, Ramprasad and Wallace (Ramprasad and Wallace, 2024) show that automatic scores overestimate LLM performance in specific domains. Consequently, a cohesive and model-agnostic framework for text verification remains unsurveyed. This paper fills that gap.

1.1. Motivation and scope

The rapid proliferation of large language models across high-stakes domains including healthcare, law, education, and scientific research has exposed a critical and persistent reliability gap: while individual verification techniques such as retrieval-augmented generation (Lewis et al., 2020), factuality scoring (Min et al., 2023b), and cross-model validation (Adlakha et al., 2024) have advanced significantly in isolation, no unified framework has consolidated these fragmented advances into a single, model-agnostic, end-to-end verification pipeline. This gap is not merely technical; its consequences are societal. Hallucinated drug interactions in clinical decision support, fabricated legal citations in judicial proceedings, and invented statistics in scientific literature represent documented failure modes with real-world impact (Beauvais, 2022; Adams et al., 2023). As Broda and Strömbäck (Broda and Strömbäck, 2024) demonstrate through interdisciplinary analysis, the unchecked diffusion of AI generated misinformation systematically erodes public trust in science, journalism, and institutional authority.

Despite extensive research investment, the verification landscape remains fragmented along three dimensions. First, *methodological fragmentation*: RAG systems (Lewis et al., 2020; Shuster et al., 2021) reduce but do not eliminate hallucination, as RAGTruth (Niu et al., 2024) empirically confirms; factuality metrics such as FActScore (Min et al., 2023b), QAFactEval (Fabbri et al., 2022), and FEQA (Durmus et al., 2020) measure consistency without providing interactive correction; and cross-model validation (Shami et al., 2026; Adlakha et al., 2024) shows promise without integration into deployable pipelines. Second, *taxonomic fragmentation*: Huang et al. (2025a) and Pesaranghader and Li (2026) identify four distinct hallucination categories factual, faithful, logical, and temporal yet no existing system addresses all four simultaneously. Third, *operational fragmentation*: existing solutions are model-specific, non-interpretable, or lack automated correction, limiting their applicability in real-world deployments where model-agnosticism, explainability, and actionability are required (Bommasani et al., 2021; Thorne, 2024).

This survey is motivated by the need to resolve this three-dimensional fragmentation. Synthesizing peer-reviewed literature spanning 2016 to 2026 across eight research dimensions hallucination taxonomy and evaluation, retrieval-augmented generation, automated fact-checking, NLP foundations, truth scoring and factuality metrics, misinformation detection, trustworthy AI frameworks, and cross-model verification we provide the first comprehensive, structured synthesis of hallucination mitigation research and propose TruthGuard AI as a theoretically-driven concept implementation of the unified verification framework, developed by directly analyzing the observed patterns from the reviewed literature. The TruthGuard AI is considered as a conceptual framework in this literature survey rather than being an empirically tested one; a full empirical analysis including datasets, baselines, implementations, and ablation tests is out of the scope of the current literature survey and is noted as an important future direction (Section 9.2). The procedure of search and screening that was performed for this literature survey and its description, including databases, search strings, screening phases, and criteria for inclusion/exclusion, can be found in Section 2.

The scope of this survey is bounded as follows. We focus exclusively on *text-generating* LLMs and their hallucination failure modes; multimodal verification (Jiang et al., 2024; Kim et al., 2024; Wang et al., 2024) is discussed as a future extension but not comprehensively surveyed. We cover verification approaches applicable as *post-generation middleware* systems that assess and correct LLM outputs without requiring access to model internals consistent with the model-agnostic deployment principle articulated by Bommasani et al. (2021). This model-agnostic approach is in reference to model-parameter independence specifically; the limitations of this independence, considering the fact that verification is dependent on the output features of the generating model, are explored in Section 7. We restrict our primary empirical analysis to English-language systems and general-domain content, while identifying multilingual and domain-specific extension as a priority future direction (Panchendrarajan and Zubiaga, 2026; Panchendrarajan et al., 2025). The survey covers literature from 2016, the year of the foundational AI safety framework of Amodei et al. (2016), through 2026, encompassing the most recent advances in LLM-based verification research.

1.2. Contributions of this paper

There are numerous existing works that discuss individual aspects of hallucination detection (Huang et al., 2025a; Tonmoy et al., 2024; Pesaranghader and Li, 2026), factuality evaluation (Min et al., 2023b; Ramprasad and Wallace, 2024; Malin et al., 2025), and retrieval-augmented generation (Gupta et al., 2024; Gao et al., 2023b). However, while there is much existing literature, no survey has yet been conducted that considers the above topics in terms of hallucination taxonomy, architecture for multi-layered verification, supporting technologies, application scenarios, and open challenges. Table 1 provides an overview of important surveys related to hallucination studies and aspects covered by these surveys.The H/M/L coverage ratings provided for Table 1, and subsequently for Table 5 introduced later in this survey, indicate a consistent qualitative scoring approach uniformly applied to all surveys under investigation: a dimension is considered as having *High* coverage if the cited survey features a separate section or subsection that provides some technical consideration of the dimension under examination (e.g., describes methods, architectures, or evaluation results relating to it); *Medium* coverage if the dimension is covered as a subtopic within a broader category without any technical discussion (e.g., briefly touched upon within a relevant section, without any particular consideration devoted to it), and *Low* coverage if there are isolated references to the dimension without any integration. This methodology is carried out manually on the basis of a review of the table of contents and body of text of each cited survey; it does not rely on a quantitative analysis of citations or inter-rater reliability analysis, and this constraint is recognized in Section 9.3. Based on the above, the key contributions of this survey can be listed as follows.

- Firstly, we provide a hierarchical taxonomy for different categories of hallucinations reported in recent literature, which includes various definitions proposed by researchers, along with their causes, forms, and difficulties in detecting them.
- A review of the current approaches for hallucination detection is presented in this work. These methods include retrieval-augmented generation, factuality measures, automatic fact checking, natural language inference, multi-model verification, and techniques of self-reflection.
- This paper analyzes the technologies that allow for the development of hallucination verification systems such as transformers, dense retrieval, knowledge graphs, credibility scoring, and truth scoring techniques.
- The review will examine the major areas where the use of hallucination detection technologies is of paramount importance, including the healthcare sector, law, science, education, and information systems.
- The review will conduct a comparative study of existing verification tools, highlight the common weaknesses and deficiencies, and compile lessons learned from which future studies can benefit.

Table 1
Summary of representative surveys on hallucination detection and related domains. Coverage levels: H = High, M = Medium, L = Low, NA = Not Applicable. Rating criteria are defined in Section 1, paragraph preceding this table.

Reference	Halluci. Tax.	RAG	Fact Check.	NLP Found.	Truth Score	Misinfo.	Cross-Model	Remarks
Huang et al. (2025a)	H	L	L	M	M	L	M	Comprehensive hallucination taxonomy; ACM TOIS
Tonmoy et al. (2024)	H	M	M	M	L	L	M	Surveys 20+ mitigation strategies
Guo et al. (2022)	L	L	H	L	L	L	L	Canonical 3-stage fact-checking pipeline
Min et al. (2023b)	M	L	M	M	H	L	L	FActScore atomic factuality evaluation
Malin et al. (2025)	M	L	M	M	H	L	M	Faithfulness metrics review; IEEE JTSP
Bommasani et al. (2021)	L	L	L	M	L	M	H	Foundation model risks; societal implications
Adlakha et al. (2024)	M	L	M	M	M	L	L	Cross-model evaluation; correctness/faithfulness
Pesaranghader and Li (2026)	H	M	M	M	M	M	M	Four-category hallucination detection taxonomy
This survey	**H**	**H**	**H**	**H**	**H**	**H**	**H**	**Comprehensive multi-dimension survey, 2016–2026**

- We identify some open problems and avenues for future work related to scaling, domain adaptation, multilingualism, real-time validation, and privacy-preserving systems in the development of trustworthy artificial intelligence.

1.3. Paper organization

The rest of this survey paper is structured as follows. The systematic review methodology adopted to identify, screen, and select the literature synthesized in this survey is described in Section 2. Definitions and background information on AI hallucinations, fact verification, trustworthy AI, and reliability testing will be covered in Section 3. State-of-the-art literature on hallucination detection and correction will be reviewed in Section 4. Applications where hallucination detection is crucial will be discussed in Section 5. Enabling technologies used for the verification process will be covered in Section 6. Comparative analysis of different approaches is provided in Section 7. Open problems in the area are addressed in Section 8. Lessons learned and possible research directions are provided in Section 9. This survey ends with conclusions

2. Review methodology

The transparency and reproducibility of the literature synthesis conducted in this survey require that the systematic process of identification, screening, and selection of the studies discussed in Sections 3–9 be reported, following the staged process represented in Fig. 1.

2.1. Preliminary investigation

This is where the systematic literature synthesis presented in this survey took its start. A preliminary investigation was made first in order to gain clarity on the scope of the search and focus the scope of the review before the systematic screening. The preliminary search was conducted using Google Scholar with the key search term phrase “hallucination detection generative AI”, as well as two or more additional terms directly related to it, for example, “factuality”, “retrieval-augmented generation”, and “fact-checking”.

After finding an initial set of relevant papers, additional keywords were identified based on their abstracts, and the bibliographic databases likely to index the target literature were found. Thereafter, the search was extended to five bibliographic databases – Scopus, Web of Science, IEEE Xplore, the ACM Digital Library, and the ACL Anthology – complemented with searches on arXiv and citation chaining using Google Scholar for recent preprints not yet indexed elsewhere. Search dates were limited to January 2016 – the date of publication of the seminal AI safety framework paper by Amodei et al. (2016) – through 30 June 2026.

Search queries were built using Boolean operators and consisted of three concept groups, for example: (“hallucination” OR “factuality” OR “factual consistency”) AND (“large language model” OR “LLM” OR “generative AI” OR “natural language generation”) AND (“detection” OR “verification” OR “fact-checking” OR “retrieval-augmented generation” OR “mitigation” OR “correction”), adapted to the field tagging and truncation features of each database.

2.2. Conducting research

Database search was chosen as the primary search strategy due to the sheer number of papers and speed of publication in this domain making it impossible to rely solely on manual searching; citation chaining using Google Scholar supplemented the structured searches with recent preprints not yet indexed elsewhere. The following inclusion and exclusion criteria were used uniformly in screening all the records found in the database searches conducted according to Section 2.1.

Inclusion: The records were included if they (i) were peer-reviewed journal articles, peer-reviewed conference papers, or official technical/model reports from major AI laboratories; (ii) were written in English; (iii) were directly addressing hallucination detection, verification, mitigation, or correction in generative AI or large language models, or were directly addressing a foundational NLP, retrieval, or evaluation technique that the surveyed hallucination literature relied upon; and (iv) were published between January 2016 and the search cut-off date.

Exclusion: The records were excluded if they were (i) non-peer-reviewed blog posts, preprints without a peer-reviewed or officially released technical report, or opinion pieces; (ii) duplicates across databases; (iii) studies where the full text could not be accessed; or (iv) studies unrelated to text-generating LLMs or to the technologies, application domains, and evaluation frameworks covered in this survey.

Screening was carried out in two steps: (1) title-and-abstract screening against the criteria above, followed by (2) full-text screening of the screened-out records. Differences in screening decisions were solved through discussion within the author team until reaching agreement.

2.3. Examining publications

Examining publications for their relevance was a two-step process. First, an initial pool of approximately 600 candidate records was identified in the database and citation chaining searches performed according to Section 2.1. Second, the relevant records were examined and selected as those significant in relation to the research aims: title-and-abstract screening against the inclusion and exclusion criteria of Section 2.2 reduced the initial pool to approximately 300 records selected for full-text screening. In relation to their relevance, the record was selected in the review only if the topic of hallucination detection, verification, mitigation, or correction was the central methodological focus of this paper, or if the paper addressed a foundational NLP, retrieval, or evaluation technique that the surveyed hallucination literature relied upon. Specifically, the record was retained if it (i) presented a new method of

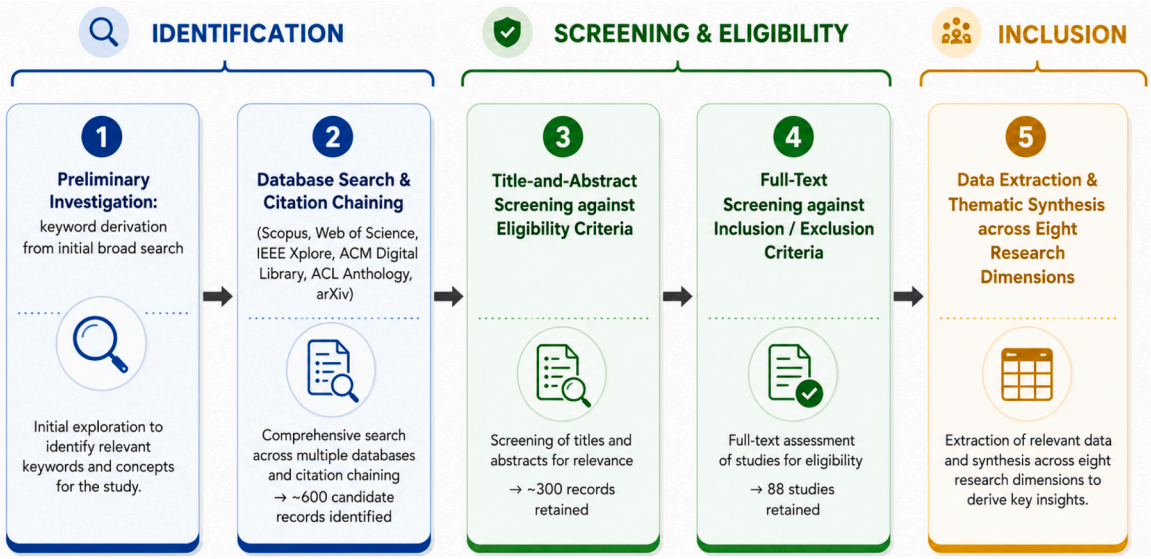


Fig. 1. Systematic process followed to identify, screen, and select the literature synthesized in this survey.

hallucination detection, verification, or correction; (ii) proposed a new extension or a new variant of an existing verification approach, for example, RAG, NLI entailment scoring, knowledge graph validation, or automated fact-checking; (iii) performed a comparative study involving one or more hallucination verification systems; or (iv) presented a foundational NLP, retrieval, or evaluation technique underlying the surveyed literature. The record was excluded if it mentioned hallucination or factuality superficially, but had no methodological contribution to make in relation to it. The result of full-text screening against the inclusion and exclusion criteria of Section 2.2 was 90 studies, which form the literature synthesized in Sections 3–9 and correspond to the reference list of the manuscript.

2.4. Extracting data

In order to fulfill the research aims of this survey, a structured spreadsheet was made to document several attributes of each of the 90 retained publications: the reference number, title, authors, publication year, publication venue, publication type (journal, conference proceeding, or official technical report), the research dimension addressed (one of the eight dimensions introduced in Section 1: hallucination taxonomy and evaluation, retrieval-augmented generation, automated fact-checking, NLP foundations, truth scoring and factuality metrics, misinformation detection, trustworthy AI frameworks, or cross-model verification), and, if applicable, the pipeline stage(s) addressed (detection, verification, mitigation, or correction, as explained in Section 3.3). Within each dimension, studies were synthesized narratively, and comparative tables were made to show the coverage of each dimension across studies (see Section 1.2 for the scoring protocol used in these tables). No formal meta-analysis or risk-of-bias score (e.g., GRADE, Cochrane RoB) was done, being consistent with the qualitative and synthesis nature of this survey; this is discussed in more detail in Section 9.3. The full list of the studies used in the qualitative synthesis corresponds to the reference list of the manuscript; given the number of entries in this list (less than 100), no separate table of inclusions is provided apart from the qualitative flow of studies selection reported in Section 2.3. consistent with common practice for narrative (as opposed to systematic) surveys of this scope.

3. Background and definitions

This section reviews the foundational concepts that underpin the hallucination verification literature. Similar to how the development of industrial standards involved agreeing on terminology before the creation of technology to enable the application (Huang et al., 2025a), the AI hallucination community has devoted considerable effort to creating precise definitions. Definitions for AI hallucinations, fact verification, trustworthy AI, and reliability verification are discussed from various viewpoints provided by practitioners in the field.

3.1. AI hallucinations: definitions and taxonomy

The concept of *hallucination* with regard to natural language generation can be defined as the generation of something which has no basis or even contradicts what has been provided from knowledge or input sources. Various definitions have been proposed in the literature.

Definition 1. Huang et al. (2025a) describe hallucination as any generated text that does not align with real-life facts, contradicts the given context, or is self-inconsistent in its generated output. This classification system classifies hallucinations according to three parameters, namely, factuality, faithfulness, and consistency, with the underlying reasons being training data errors and attention calibration failure.

Definition 2. According to Tonmoy et al. (2024), hallucinations refer to LLMs' behavior of making false but confident claims which are either impossible to verify or are contradicted by known information. They classify more than twenty examples of such behavior depending on tasks and models used.

Definition 3. According to Pesaranghader and Li (2026), there are four categories of detections fact, faithful, logic, and time hallucination, and each of which needs to be tackled differently.

Definition 4. According to Anh-Hoang et al. (2025), hallucinations can be understood as output varying systematically based on prompting approach but not based on the actual facts, highlighting the impact of how the input is formulated on the rate of hallucinations generated in different types of tasks.

Based on the above definitions, it becomes clear that hallucination is a multidimensional concept consisting of, at least, four kinds of errors: *factual hallucination*, where an assertion is contradicted by world facts; *faithful hallucination*, where an assertion is inconsistent with the provided input; *logical hallucination*, where an assertion contradicts itself; and *temporal hallucination*, where date information is inaccurate. Table 2 shows a taxonomy of these types of hallucinations along with their definitions, sample examples, and detection methods.

Table 2
Taxonomy of hallucination types in generative AI, with definitions, examples, and primary detection approaches documented in recent literature.

Type	Definition	Example	Primary Detection Approach
Factual Hallucination	Claim contradicted by verifiable world knowledge	Stating a historical figure was born in the wrong country	RAG retrieval (Lewis et al., 2020; Asai et al., 2023); atomic factuality scoring (Min et al., 2023b)
Faithful Hallucination	Claim inconsistent with provided input context	Summarizing a document as stating X when it states Y	NLI entailment scoring (Fabbri et al., 2022; Durmus et al., 2020); faithfulness metrics (Lamsiyah et al., 2025; Malin et al., 2025)
Logical Hallucination	Internally contradictory or self-inconsistent claim	Asserting an event preceded its own stated preconditions	Consistency checking (Wang et al., 2020; Song et al., 2025); NLI contradiction scoring (He et al., 2020)
Temporal Hallucination	Incorrect dates, durations, or chronological ordering	Claiming a documented event occurred in the wrong year	Temporal graph validation (Drchal et al., 2024); date extraction NER (Devlin et al., 2019)
Entity Hallucination	Invented names, organizations, or products	Citing a non-existent paper by a real author	Named entity grounding (Panchendrarajan and Zubiaga, 2026); knowledge graph validation (Shami et al., 2026)

3.2. Differences between hallucination and related terms

The terms *hallucination*, *misinformation*, *factual error*, and *unfaithful generation* are often used interchangeably within the literature covered in this survey, but they denote related yet distinct concepts that require different approaches to verification.

Hallucination is a property of the *generation process* where the output of a generative model contains unsupported or contradictory to training knowledge, input, and itself information, regardless of whether it happens to be a true claim in the real world. *Factual error* is a property of a *claim* – a false claim about verifiable knowledge of the real world, unrelated to its source. *Misinformation* – as addressed in the misinformation detection literature surveyed in Section 4 – is a property of *circulating content*, regardless of its origin and intent, and includes disinformation, human factual errors, and AI hallucinations, among others. Misinformation detection is thus a superset problem that cannot be solved solely with the help of hallucination verification techniques. *Unfaithful generation* is, in turn, a property of *context-grounded generation tasks* such as summarization or RAG: the output is considered unfaithful if it contradicts the given source or evidence, even if it happens to be a factually correct claim about the world – i.e., a faithful summary can reproduce a factual error from its source without being unfaithful, while an unfaithful summary can include a factually correct claim not present in the source. The primary concern of this survey, in accordance with the definition in Section 1, is hallucination in the former sense (a generation process property), while factual error and unfaithful generation verification serve as operational measures for this property, with the misinformation detection being a related adjacent field with useful credibility assessment techniques (Section 3.4).

3.3. Detection, verification, mitigation, and correction: definition of pipeline stages

The terms *detection*, *verification*, *mitigation*, and *correction* are used together in many studies about hallucination, but they correspond to distinct stages of the verification process each having a unique goal, input, and output. This survey employs the following explicit definitions consistently in the remaining part of the paper.

Detection is an identification of the generated content *suspected* to be a hallucination, but not confirmed or disproven against evidence. Detection usually involves the analysis of internal or external generation process signals, such as model uncertainty, consistency of multiple outputs (Manakul et al., 2023), or consistency with a retrieved passage, and generates a flag or confidence score rather than a definitive judgement.

Verification is a process of checking of a flagged (or unflagged) claim against the evidence and determining if it is supported, refuted, or unverifiable. Verification extends directly the fact-checking pipeline proposed by Guo et al. (2022) (claim detection, evidence retrieval, verdict prediction) and performs an evidence-grounded evaluation of the claim, usually with the help of NLI entailment scoring (Fabbri et al., 2022; Durmus et al., 2020; He et al., 2020) or knowledge-graph validation (Wang et al., 2020; Shami et al., 2026).

Mitigation refers to the interventions made *during or before the generation process* that reduces the likelihood of hallucinating in the first place, rather than checking the content afterwards. RAG (Lewis et al., 2020; Asai et al., 2023), instruction tuning with RLHF (Ouyang et al., 2022), and DPO-based preference alignment (Rafailov et al., 2023) are examples of mitigation techniques – they lower the base rate of hallucination but do not prevent hallucination in any particular output (Niu et al., 2024).

Correction is the *post-generation* re-writing of the output to eliminate unsupported claims, once they have been flagged and verified. The RARR framework (Gao et al., 2023a) exemplifies a correction technique that retrieves evidence for or against a generated claim and re-writes it accordingly.

Table 3 shows how individual techniques covered in this survey are mapped to the pipeline stage(s) they solve, and demonstrates that most of them address one or two of the four stages only, which is the reason for the multi-layer architectures described in Section 7.

3.4. Fact verification

Fact-checking, defined as the task of determining whether a certain claim can be substantiated, rejected, or neither based on evidence, has been investigated intensively within the NLP literature before the advent of LLMs. Guo et al. (2022) set up the gold-standard three-step pipeline for automatic fact-checking: detecting claims, retrieving evidence, and making judgment predictions. This pipeline has motivated a number of hallucination verification architectures (Schlichtkrull et al., 2023; Putta et al., 2025; Gong et al., 2026).

Some of the research on extending the fundamental pipeline is as follows. Schlichtkrull et al. (2023) created AVerITeC, which is a dataset for real-world claim verification through multi-hop web evidence retrieval and sets the benchmark for testing the end-to-end performance of verification systems. Akhtar et al. (2024) propose Ev2r metric for systematically analyzing the quality of evidence used to make verdicts during automated fact-checking. Gong et al. (2026) extend the paradigm by adding multiple sources and agents for evidence retrieval, showing that spreading the retrieval task among agents increases diversity and lowers biases of retrieval processes. Eldifrawi et al. (2024) address the problem of automated justification for claims by arguing that fact-checking without a verifiable justification has limited practical value.

The issue of evidential verification has also received attention in relation to source credibility. According to Chrysidis et al. (2024), reliable

Table 3
Mapping of representative hallucination-related methods discussed in this survey to the detection, verification, mitigation, and correction pipeline stages.

Method / System	Detection	Verification	Mitigation	Correction	Key Reference
Standard RAG			✓		Lewis et al. (2020)
Self-RAG	✓		✓		Asai et al. (2023)
RAGAS	✓	✓			Es et al. (2024)
Three-step fact-checking pipeline		✓			Guo et al. (2022)
AVerITeC		✓			Schlichtkrull et al. (2023)
SelfCheckGPT	✓				Manakul et al. (2023)
RARR		✓		✓	Gao et al. (2023a)
NLI entailment scoring (DeBERTa)		✓			He et al. (2020)
Knowledge-graph validation		✓			Shami et al. (2026)
Instruction tuning / RLHF / DPO			✓		Ouyang et al. (2022), Rafailov et al. (2023)

source information, indicating credible, unreliable or leaked evidence, greatly impacts the accuracy of judgement and therefore warrants the consideration of source authority rating as part of verification processes. Srba et al. (2026), offer an extensive survey of automatically assessing credibility from textual information in the era of LLMs, providing the necessary theoretical foundation for domain authority rating as part of multi-layer verification. Panchendrarajan and Zubiaga (2026), extend claim detection to cross-lingual contexts via entity-based approaches, showing that successful hallucination detection must consider multilinguality.

Read together, these works reflect two complementary rather than competing conceptions of fact verification. The first, exemplified by Guo et al. (2022) and its extensions (Schlichtkrull et al., 2023; Gong et al., 2026; Akhtar et al., 2024), treats verification as a *pipeline problem*: a claim is detected, evidence is retrieved, and a verdict is predicted, with recent work refining the quality of each stage (multi-hop retrieval (Schlichtkrull et al., 2023), evidence-quality scoring (Akhtar et al., 2024), multi-agent retrieval (Gong et al., 2026)) or adding a justification requirement (Eldifrawi et al., 2024). The second, exemplified by Chrysidis et al. (2024) and Srba et al. (2026), treats verification as a *source-credibility problem*: the reliability of a verdict depends on the trustworthiness of the evidence source, independent of how that evidence was retrieved. These are not mutually exclusive framings; a source-credibility signal can be incorporated as an additional weighting factor within the verdict-prediction stage of the pipeline view. This survey adopts the pipeline view of Guo et al. (2022) as the structural backbone for the *verification* stage defined in Section 3.3, augmented with source-credibility weighting (Chrysidis et al., 2024; Srba et al., 2026) and cross-lingual claim detection (Panchendrarajan and Zubiaga, 2026) as refinements within that pipeline, rather than as a separate competing framework. This adopted framing is used consistently in the comparative analysis of Section 7.

3.5. Trustworthy AI

The concept of trustworthy AI encompasses reliability, interpretability, fairness, safety, and alignment with human values as fundamental design requirements for AI systems (Amodei et al., 2016; Bommasani et al., 2021; Weidinger et al., 2021). For generative AI specifically, verifiable hallucination detection is a necessary condition for achieving such trustworthy systems.

The paper by Bommasani et al. (2021) discusses how foundation models create social-scale propagation of training errors, and therefore calls for external verification middleware as a necessary safety layer. Weidinger et al. (2021) consider ethical and social implications of LLMs on several levels of potential harms, including generation of misleading content, thus showing the ethical necessity of hallucination mitigation. Thorne Thorne (2024) establishes that interpretability is a key prerequisite for correct user trust estimation, arguing that a single aggregate score is insufficient for users to make reasonable decisions on AI-generated content. Jeong et al. (2025) propose a framework for analyzing ethics and trustworthiness of generative AI systems based on several critical aspects including factual accuracy, transparency, fairness, and safety.

Principles on which Amodei et al. (2016) base their work on the topic of AI safety can be applied to the research on hallucination verification since they discuss issues connected to side effects of AI, in case if the system is working under misaligned confidence. Hendrycks et al. (2020) explore more general topic, concerning issues connected to the alignment of AI with common human values, thus, hallucination being one of the problems connected with such an alignment.

The above perspectives of trustworthy AI can be categorized into three different but intersecting framings rather than a single definition. The *societal-risk framing* of Bommasani et al. (2021) and Weidinger et al. (2021) sees trustworthiness as an attribute of deployment at scale, focusing on the risks a foundation model can cause through the propagation of the errors it makes. The *interpretability framing* of Thorne (2024) and Jeong et al. (2025) regards trustworthiness as an attribute of the verification output, focusing on the question of whether a user is able to comprehend and calibrate their trust in the verification result appropriately. The *alignment framing* of Amodei et al. (2016) and Hendrycks et al. (2020) sees trustworthiness as an attribute of the training objective of the model, asking whether the behavior of the system reflects the human values intended. In this survey, none of the above framings are chosen exclusively; on the contrary, the approach is to view hallucination verification as an essential but alone insufficient element of trustworthy AI, while choosing the interpretability framing of Thorne (2024) specifically for verification *output* (Section 8) and the societal-risk framing of Bommasani et al. (2021) for the argument made in this survey about model-agnostic verification middleware (Section 1).

3.6. Reliability assessment

Reliability assessment in the context of LLM outputs refers to the systematic quantification of the degree to which generated content can be trusted to be factually accurate and contextually appropriate. Several approaches have been proposed in the literature.

As shown by Ramprasad and Wallace (2024), automatic factuality scoring tends to overestimate the accuracy of language models in domain-specific scenarios, which underscores the limitations of reliability estimation using only one type of metrics in contrast to human assessments of actual accuracy. Zhang et al. (2024a) also find discrepancies between automated measures of faithfulness and human assessments of citation accuracy, pointing to the need for composite multi-signal reliability scoring. Finally, FACTS Grounding Leaderboard created by Jacovi et al. (2025) provides a comparative evaluation framework that tests the ability of LLMs to ground their responses to long-form inputs. In their thorough review of faithfulness measures for assessing hallucinations in LLMs, published in IEEE Journal of Selected Topics in Signal Processing, Malin et al. (2025) conclude that there is no single metric capable of capturing all aspects of factual accuracy, advocating for multi-faithfulness evaluation methods.

4. State-of-the-art literature review

This section provides a comprehensive review of the principal research directions in hallucination detection, evaluation, and mitigation. This analysis is conducted across eight dimensions, which correspond to the methodological lines within this field. They include: hallucination taxonomy and evaluation, retrieval-augmented generation, automatic fact-checking, foundations of NLP, truth scoring and metrics, misinformation detection, cross-model and knowledge graph verification, and self-verification.

While Section 3 provided the conceptual framework of terms and taxonomies utilized in this survey (the nature of hallucination, the meaning of verification and trustworthy AI, and the pipeline phase that the method targets), this section provides the synthesis of the relevant *methodology* literature in each of the eight dimensions: the particular methodologies, benchmarks, and systems that embody the previously defined concepts, in chronological order within each dimension to illustrate the development of methodology. If the subsections below need to revisit a term introduced in Section 3 (such as hallucination taxonomy and the definition of trustworthy AI), then the discussion is restricted to the methodology-related content not provided in the latter section.

4.1. Hallucination taxonomy and evaluation

The taxonomic categories underlying this subsection – factual, faithful, logical, and temporal hallucination – were defined and compared in Section 3.1; this subsection does not restate those definitions but instead traces the chronological development of the benchmarks used to operationalize them. Benchmark development became an important step towards better understanding and assessing hallucinations. Specifically, TruthfulQA (Lin et al., 2022) created the first benchmark by identifying factual failures in high-performing LLMs, while FActScore (Min et al., 2023b) developed the method of atomic claim decomposition for precise factual evaluation. QAFactEval (Fabbri et al., 2022) and FEQA (Durmus et al., 2020) were used to assess the consistency of LLM-generated summaries in QA paradigm. However, an important follow-up study by Ramprasad and Wallace (2024), Zhang et al. (2024a) showed that automatic metrics overestimated performance by multiple metrics compared to human evaluation. This issue was addressed by several composite multi-metric evaluation approaches suggested by Malin et al. (2025), Lamsiyah et al. (2025), and the FACTS Grounding Leaderboard (Jacovi et al., 2025).

The approximate sizes reported above reflect the authors' familiarity with these widely cited benchmarks as documented in their original publications. Table 4 makes clear that no single benchmark jointly covers multiple languages, multiple domains, and end-to-end (detection-through-correction) evaluation: English-only, single-domain benchmarks such as FActScore, QAFactEval, and FEQA dominate the landscape, multilingual coverage is confined to claim-clustering resources such as MultiClaimNet rather than full verification benchmarks, and evaluation strategies range from human span annotation (RAGTruth) to LLM-judge scoring (FACTS Grounding), which are not directly comparable to one another. This fragmentation reinforces the case, developed further in Section 8, for benchmark suites that jointly vary language, domain, and pipeline stage rather than treating each as an independent axis.

4.2. Retrieval-augmented generation

Retrieval-augmented generation (RAG) became one of the most active areas in developing approaches for mitigating hallucinations. The basic design of the RAG architecture has been proposed by Lewis et al. (2020) in NeurIPS 2020. They discovered that conditioning generation on external retrieval passages greatly reduced hallucinations in knowledge-intensive tasks, a result corroborated by Shuster et al. (2021) for conversation scenarios. Further developments in RAG models included Self-RAG (Asai et al., 2023), which allows adaptive retrieval through self-critiquing using reflection tokens, and Search-in-the-Chain (Xu et al., 2024), which refines answers through iterative retrieval. A novel framework, RAGAS (Es et al., 2024), allows automatic evaluation of RAG pipelines on different dimensions including faithfulness, answer relevance, and retrieval recall. Moreover, recent studies have provided an extensive overview of RAG architectures; e.g., those presented by Gao et al. (2023b) and (Gupta et al., 2024). Despite these improvements, however, RAG systems are not yet fully reliable. Niu et al. (2024) proves empirically that even well-structured RAG pipelines may hallucinate if the retrieved context is inadequate or ambiguous. Wang et al. (2023) prove that self-knowledge estimation helps reduce unnecessary retrieval costs, whereas (Manakul et al., 2023) proposes a zero-resource detection of hallucinations using stochastic self-sampling. Moreover, the RARR framework (Gao et al., 2023a) aims at correcting hallucinated claims after generation through context retrieval. Mallen et al. (2023) have demonstrated that parametric and non-parametric models are vulnerable to long-tail factual queries, and Shi et al. (2023) have proven that even if factual information is present, LLMs are easily distracted by irrelevant context.

4.2.1. Traditional RAG versus agentic RAG

The RAG variations mentioned above – standard RAG (Lewis et al., 2020), Self-RAG (Asai et al., 2023), RAGAS (Es et al., 2024), and RAGTruth (Niu et al., 2024) – are all largely examples of what *traditional* retrieval entails: a static evidence set conditioned via a single pass of retrieval is the basis for generation with no room for re-planning of retrieval operations in light of intermediate reasoning failures. The difference between the *traditional* and *agentic* RAG is clearly articulated by Neha and Bhati Neha and Bhati (2025), where agentic RAG means autonomous agent(s) planning and conducting retrieval operations, invoking tools, reiterating retrieval in light of intermediate outputs, and reflecting on partial results before giving the final output. When it comes to verification of hallucinations, agentic RAG becomes especially relevant due to the fact that, unlike in the traditional setting, evidence gathering here is a process that involves multiple retrieval stages and is not reduced to a single lookup operation: agents may ask for more evidence when needed, divide the claim into smaller parts and verify those separately, invoke specialized tools like calculators, knowledge graph searches, or citation checkers during reflection before giving the final verdict (Neha and Bhati, 2025). The Search-In-The-Chain framework (Xu et al., 2024) and the multi-agent evidence-retrieval system by Gong et al. (2026) may both be considered an early example of agentic RAG in the context of hallucination verification research, despite the fact that agentic RAG was not explicitly stated. Agentic RAG should be preferred over traditional one when complex multi-hop reasoning is desired while the latter works better in terms of latency and simplicity; however, agentic RAG requires more inference calls (Neha and Bhati, 2025), which directly affects our scalability discussions in Section 8.

4.3. Automated fact-checking

Automated fact-checking is a critical component of the evidence-based methodology for hallucination verification. The foundational three-step pipeline of claim detection, evidence retrieval, and verdict prediction was first proposed by Guo et al. (2022). This framework has subsequently led to a plethora of designs for verifying claims with hallucinations.Schlichtkrull et al. (2023) expanded on this paradigm to address the problem of multi-hop claim verification on web-sourced evidence, whereas (Putta et al., 2025) and Gong et al. (2026) advanced the field with web-sourced and multi-agent heterogeneous source methods for evidence retrieval respectively. The Akhtar et al. (2024) identified a critical gap by evaluating the evidence retrieval phase in the fact-checking pipeline. Recent research has enhanced and broadened the

Table 4
Benchmark datasets for hallucination detection and verification research, with scope, approximate size, language coverage, domain, annotation strategy, evaluation metric, and key limitations.

Benchmark (Ref., Year)	Size (approx.)	Scope / Task Type	Language(s)	Domain	Annotation Strategy, Metric & Key Limitation
TruthfulQA (Lin et al., 2022),	~800 questions	Factual failure detection in high-performing LLMs	English	Open-domain, adversarially selected topics	Human-authored adversarial questions; human/automated truthful + informative judgment; small scale limits statistical power and its adversarial framing may not generalize to naturalistic prompts
FActScore (Min et al., 2023b),	~500 generations	Atomic-level factual precision in long-form generation	English	Biography generation	Atomic-fact decomposition validated against Wikipedia; FActScore (% supported atomic facts); reliant on Wikipedia as ground truth and limited to the biography domain
QAFactEval (Fabbri et al., 2022),	Standard summarization test sets (e.g., CNN/DM, XSum)	QA-based factual consistency in summarization	English	News summarization	Automatically generated QA pairs; QA-answer-overlap score; summarization-specific and sensitive to question-generation quality
FEQA (Durmus et al., 2020),	Standard summarization test sets	Faithfulness in abstractive summarization	English	News summarization	QG/QA-based answer-overlap F1; summarization-only scope and dependent on QG model quality
AVerITeC (Schlichtkrull et al., 2023),	~4,500 real-world claims	Real-world multi-hop claim verification via web evidence	English	Open-domain, fact-checked news claims	Claims sourced from fact-checking sites with human-collected justifications; question-answer and verdict match; noisy, fast-changing web evidence complicates reproducibility
RAGTruth (Niu et al., 2024),	~18,000 response-level annotations	Hallucination corpus for RAG system evaluation	English	QA, summarization, data-to-text (RAG settings)	Human span-level hallucination annotation; span-level precision/recall; single-language and RAG-specific, may not generalize to non-RAG generation
FACTS Grounding (Jacovi et al., 2025),	Leaderboard-scale long-form responses	Source-grounding accuracy for long-form LLM responses	English	Long-form, document-grounded generation	LLM-judge-based grounding scoring; grounding accuracy score; reliance on LLM judges raises circularity concerns for LLM-generated text
MultiClaim-Net (Panchendrarajan et al., 2025),	Large-scale multilingual claim clusters	Multilingual fact-checked claim cluster evaluation	Multiple languages	Open-domain fact-checked claims	Cluster-based claim aggregation; clustering-quality metrics; language coverage is uneven across resource-rich vs. low-resource languages
Ev2r (Akhtar et al., 2024),	Evidence sets drawn from existing fact-checking corpora	Evidence retrieval quality in fact-checking pipelines	English	Fact-checking evidence retrieval	Reference-based evidence-quality scoring; Ev2r score; narrowly scoped to the evidence-retrieval stage rather than end-to-end verification
GO FIGURE (Gabriel et al., 2021),	Meta-evaluation over multiple summarization datasets	Meta-evaluation of factuality metrics in summarization	English	News summarization	Correlation with human factuality judgments; meta-evaluation correlation score; evaluates metrics rather than generation systems directly

scope of automated fact-checking. For example, the ability to discriminate between credible, unreliable, and leaked sources greatly improved verdict performance in the study conducted by Chrysidis et al. (2024), whereas the importance of automated justification generation along with verdict was noted by Eldifrawi et al. (2024). Panchendrarajan and Zubiaga (2026) introduced entity-aware methods for claim detection in cross-lingual settings, and Panchendrarajan et al. (2025) contributed the first massively multilingual benchmark for fact-checked claim clusters. Drchal et al. (2024) designed a language-agnostic fact-checking pipeline, whereas (Sahitaj et al., 2025) demonstrated that large LLMs could generate and evaluate fact-checking rationales with appropriate prompts.

4.4. Natural language processing foundations

The NLP principles underlying the technology to detect hallucinations rely on the transformer model designed by Vaswani et al. (2017), with its multi-headed self-attention feature which makes the model capable of capturing long-term dependencies necessary to decompose the claim and assess the implications. Bidirectional pre-training was shown to be important for fine-tuning BERT (Devlin et al., 2019) for NLI and NER; Liu et al. (2019) enhanced pre-training procedures while He et al. (2020), with its disentangled attention mechanism, is currently the best-performing model for detecting entailments. Larger generative models such as Brown et al. (2020), Achiam et al. (2023), Touvron et al. (2023) play a dual role as sources of hallucinations in need of verification and references in cross-model consensus approaches (Adlakha et al., 2024). Finally, certain training practices were reported to impact hallucination rates within large language models. For instance, instruction tuning with RLHF (Ouyang et al., 2022) as well as direct optimization of preferences (Rafailov et al., 2023) lead to lower hallucination frequencies via preference alignment. Chain-of-thought prompting (Wei et al., 2022b) contributes to better structured multi-step reasoning and thus to more reliable fact-checking of complicated statements. The HELM evaluation system (Liang et al., 2022) offers a holistic view of the evaluation process by integrating factuality among other dimensions. (Min et al., 2023a; Zhao et al., 2023) survey pre-training progress relevant for verification applications.

Moreover, the exact architectural features and alignments that have been designed into each LLM family also affect how likely the model will be to hallucinate and how suitable it is for hallucination verification purposes. Bhati et al. (2026) conduct a comparative survey of the major architectures of decoder-only models such as GPT, LLaMA 2, Gemini, Claude, DeepSeek, Falcon, and Qwen, systematically analyzing

such aspects as attention mechanics (e.g., multi-head, grouped-query and multi-head latent attention), normalization techniques, mixture-of-experts routing, long-context modeling, as well as different alignment pipelines like instruction tuning, RLHF, DPO, and Constitutional AI. According to the researchers' analysis, deeper alignment pipelines (e.g., Constitutional AI and RLAIF) typically contradict attention-efficiency optimization, while the figures of benchmark performance reported by model vendors usually differ from those that can be independently replicated by several percentage points (Bhati et al., 2026). The relevance of these conclusions to hallucination verification is twofold: first, the results suggest that alignment technique in itself is not enough to predict factual reliability of the model since efficiency-related and alignment-related design choices tend to overlap at the architectural level; second, self-reported benchmark performance, including factuality-related benchmarks, cannot always be trusted as a measure of hallucination rate and, therefore, reinforces the argument made throughout the entire paper regarding the importance of using external verification middleware rather than just evaluating vendor-provided evaluation (Bommasani et al., 2021).

4.5. Misinformation detection

The literature on misinformation detection and verification includes numerous methodological resources and conceptual frames that can be applied directly to the problem of hallucination detection. Srba et al. (2026) present an extensive overview of automatic credibility assessment based on textual data which provides solid theoretical ground for the development of multi-layer credibility verification systems. Nguea-jio et al. (2025) experimentally show that transformer models always outperform traditional machine learning algorithms in the problem of detecting fakes. In turn, Raza et al. (2025), Qazi et al. (2025) generalize this result to various languages and corpora, proving that evidence-based retrieval works better than generation in cases of large-scale fake news detection. Beer and Matthee (2020), Thibault et al. (2025) offer overviews of detection methodology for the evaluation of credibility. Sociological and psychological aspects of misinformation are also relevant for developing the verification system. Automation bias documented by Beauvais (2022), Adams et al. (2023) causes users, especially non-experts within a particular domain, to blindly trust algorithmic outputs, making the interpretability feature crucial for credibility verification. Broda and Strömbäck (2024) provide interdisciplinarity-supported evidence that AI-based misinformation systematically decreases trust within institutions. Domain-related and regional examples of AI-based fake content spread are provided by Tomassi et al. (2025), Kishwar and Zafar (2023), respectively. Multimodal applications of AI-based fakes in a cloud environment are studied by Wang et al. (2024), Abduljaleel and Ali (2024), Cavus et al. (2024).

Misinformation-detection models presented in the preceding paragraphs either separate the two tasks into retrieval and fake-news classification or consider them separately for RAG and classification. Systems that integrate the fact-checking process based on retrieval-augmented generation architecture provide a closer integration of these two tasks. For instance, TrumorGPT (Hang et al., 2025) proposes to combine a language model with graph-based retrieval-augmented generation (GraphRAG) on constantly refreshed semantic health knowledge graphs for fact-checking of health claims, tackling the problem of hallucinations at the fact-checking stage rather than only considering article classification as fake or real. Similarly, Bai and Fu (2024) present FCRV (Full-Context Retrieval and Verification), in which a fact-checking task is integrated through LLM-based extraction of claims and evidence retrieval by RAG to build "full-context" representation of the news article prior to its verification. These models demonstrate that for the hallucination fact-checking problem in particular, a coupled pipeline approach proposed in Section 4.2.1 is more appropriate than approaches from the mostly classification-oriented misinformation detection literature above.

4.6. Cross-model verification and knowledge graph approaches

The technique of cross-model verification relies on the idea that independent models cannot hallucinate in the same way; thus, their inter-model agreement is an efficient verification cue. Adlakha et al. (2024) show empirically that the scores in cross-model evaluation align highly with human assessments of hallucination presence. Meanwhile, Manakul et al. (2023) use stochastic self-sampling during generation to detect hallucinations at zero-resource conditions, called SelfCheckGPT. Shami et al. (2026) prove the efficacy of using SPARQL-NLI methods in verifying knowledge graphs' factuality, specifically in entity-attribute consistency tasks. Knowledge graph-based techniques are beneficial additions to retrieval-based approaches as they offer symbolic and explainable paths to verification. According to Wang et al. (2020), cross-graph representation learning provides the theoretical framework behind the approach to entity-attribute consistency verification in pipelines. Huang et al. (2025b) apply a similar principle to verify factuality of a statement using graph-based fact checking and propagation of evidence along multi-hop relational chains. Song et al. (2025) achieve promising results with energy networks that perform set-consistency verification in a structured manner.

4.7. Trustworthy AI and ethical frameworks

The normative foundations of trustworthy AI relevant to hallucination verification – the societal-risk, interpretability, and alignment framings, and the framework this survey adopts from them – are established in Section 3.5 and are not repeated here. The literature-specific addition this subsection contributes is empirical rather than definitional: Perez et al. (2022) show by adversarial red-teaming experiments that even properly aligned models can be made to generate hallucinations if prompted correctly, demonstrating that the alignment framing of Section 3.5 (Amodei et al., 2016; Hendrycks et al., 2020) is necessary but not sufficient on its own, and that red-teaming of verification systems – not just of the underlying generative model – is an important complementary safeguard, a point returned to in the future-directions discussion of Section 9.2.

5. Applications of hallucination detection systems

Hallucination detection and mitigation techniques are highly relevant to many fields, where factual accuracy plays a crucial role. This section covers key application areas where such techniques have been investigated or applied, providing domain-specific details based on the existing literature. Fig. 2 depicts the key application domains for hallucination detection systems.

5.1. Healthcare and clinical decision support

Healthcare emerges as a critical domain of applications for the purpose of verifying hallucinations. Using AI content in clinical decision support, patient interactions, and literature summarization poses serious risks, especially in case of possible hallucinations. Multiple studies show that guidelines or medical information generated by LLMs contain proven hallucinations (Beauvais, 2022; Adams et al., 2023), posing a clear threat to patient health and safety.

The current body of literature states that the prevalence of hallucinations in medical domains significantly exceeds general QA scenarios for queries regarding specialized clinical knowledge (Pesaranghader and Li, 2026; Kaddour et al., 2023). Such increase can be explained by the long tail distribution of specialized knowledge compared to general pretraining datasets, as proven by Mallen et al. (2023). In their work, they showed that parametric memories cannot be considered reliable sources of knowledge within a domain underrepresented in training corpora. While RAG systems perform promisingly in specialized domains, by using the context from specialized literature (Lewis et al., 2020; Asai et al.,

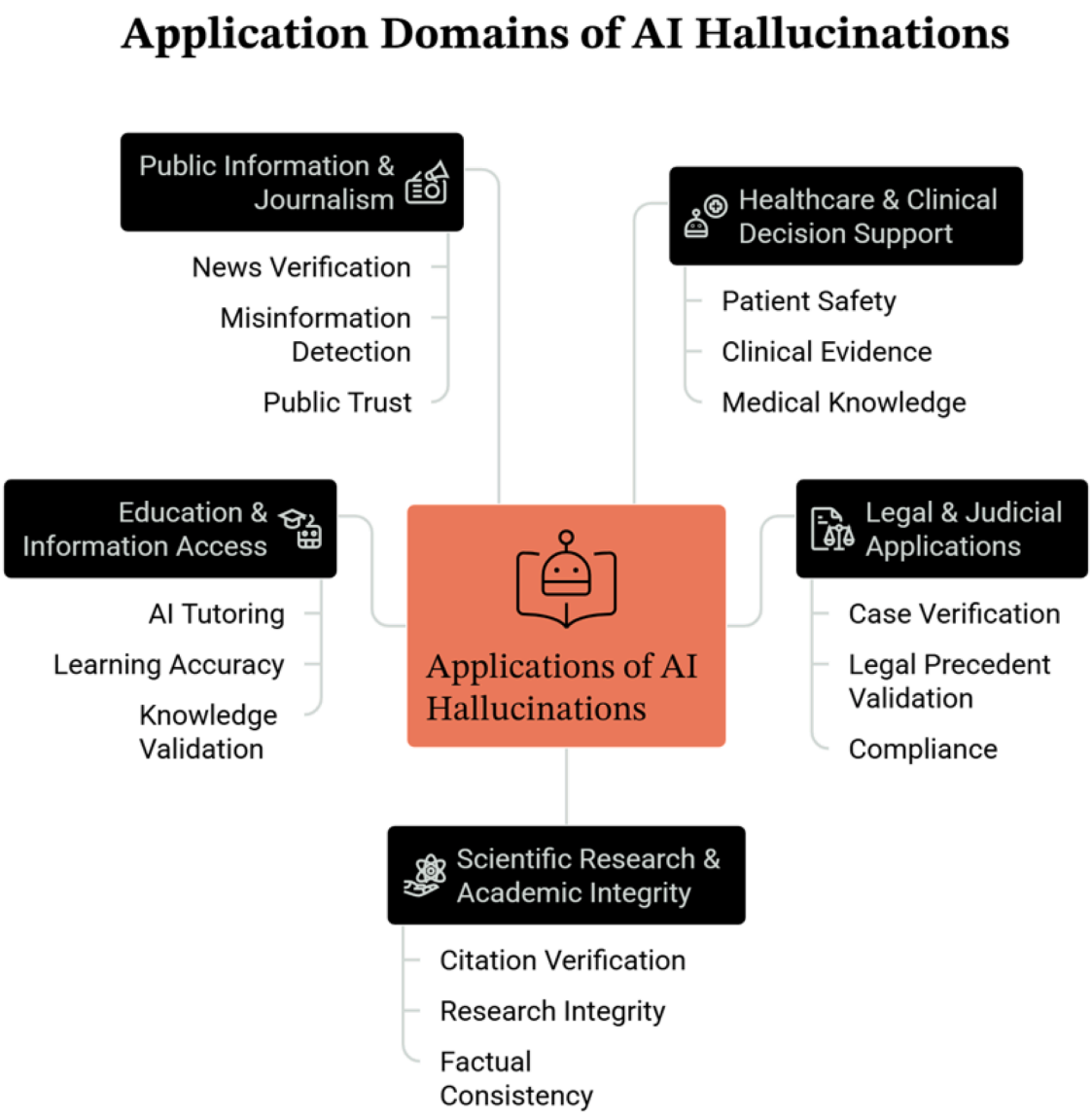


Fig. 2. Principal application domains of hallucination detection systems in generative AI. Each domain presents distinct verification requirements with respect to hallucination type, acceptable false-positive rate, evidence source requirements, and interpretability standards.

2023), RAGTruth (Niu et al., 2024) proved that the hallucinations still happen when the context is ambiguous or retrieval fails to identify all relevant evidence Trustworthy AI frameworks designed for healthcare pay special attention to model interpretability and uncertainty (Jeong et al., 2025; Thorne, 2024). Bommasani et al. (2021) highlight that societal consequences of deploying a foundation model in medicine make the external middleware verification necessary, not just internal model calibration. Finally, recent literature on multimodal verification (Jiang et al., 2024; Abduljaleel and Ali, 2024) proves that verification of AI systems used in healthcare should not be limited only to text hallucination detection but also include images, structured clinical information, and temporal records.

For healthcare specifically, the evidence sources most relevant to verification are peer-reviewed clinical literature, curated drug-interaction and dosage databases, and institutionally endorsed clinical practice guidelines, since these are the sources against which clinicians themselves would check a claim (Lewis et al., 2020; Bommasani et al., 2021). Given the potential for patient harm, the literature reviewed in this survey supports treating clinical hallucination as warranting a near-zero acceptable error rate for claims with direct treatment implications (e.g., dosage, contraindication, or diagnostic criteria), in contrast to the higher tolerance for error that may be acceptable in low-stakes, exploratory information-access settings discussed in Section 5.

5.2. *Legal and judicial applications*

The field of law also entails similarly high stakes. It has been reported that LLMs are capable of inventing cases, misattribution of legal precedents, and creating a legally valid but factually erroneous summary of a statute (Kaddour et al., 2023; Adams et al., 2023). The repercussions associated with such a phenomenon for legal practice can be significant, ranging from punishment for lawyers to miscarriages of justice during adversarial proceedings.

Current methods of verifying facts in a legal context have involved adopting the established three-step process of fact checking (Guo et al., 2022) and using ontology or knowledge graph techniques tailored to legal documents (Wang et al., 2020; Huang et al., 2025b). Shami et al. (2026) prove that knowledge graph validation techniques are highly useful for entity-attribute consistency tasks in legal texts, especially because accurate attribution of a holding to a court and date is very important. Eldifrawi et al. (2024) highlight the importance of automated

justification generation in the legal domain, as users need both a verdict on the veracity of a claim and its evidence base.

In the legal context, the evidence sources that are most relevant in terms of verification are the primary legal sources: official reporters of case law, statute and regulations databases, and court docket records, because these are the official authorities against which the claim can be verified (Wang et al., 2020; Shami et al., 2026). As a single fake citation is capable of damaging the reputation of an entire legal document and getting the individual involved in trouble, the literature presented in this survey indicates the tolerance of error in the claims about citation existence and holding attribution as equal to zero or close to zero.

5.3. Scientific research and academic integrity

In scientific research, hallucination is manifested through the inclusion of fictitious citations, incorrect attribution of statistics, and erroneous representations of experimental results (Huang et al., 2025a; Pesaranghader and Li, 2026). The rapid proliferation of LLM-based writing support tools within academia has caused a great concern regarding hallucination in scientific text, prompting institutions such as journal editors and research integrity offices to develop strategies for hallucination detection (Amodei et al., 2016; Weidinger et al., 2021).

FActScore (Min et al., 2023b) was developed for the specific purpose of measuring the precision of facts in text generation and can be used directly to verify scientific text. Muhlgay et al. (2024) propose techniques to automatically generate benchmarks for evaluating factuality in language models, creating scalable means for assessing LLMs for factual information. FACTS Grounding Leaderboard (Jacovi et al., 2025) is a benchmark for measuring an LLM's capability for grounding text based on input documents.

For scientific writing, the main sources of evidence are peer-reviewed publication databases and citation indices (such as the indexed literature on which Scopus and Web of Science rely), which allow one to verify the existence, authorship, and claims made in an artificial citation (Min et al., 2023b; Muhlgay et al., 2024). Since artificial citations and misattributions of statistics jeopardize research integrity and could contaminate the scientific record if not identified, the literature shows that there should be close to zero tolerance for such mistakes when it comes to citation existence and statistical attribution in the context of scientific writing.

5.4. Education and information access

Application areas of LLMs within education such as AI-powered tutoring systems, automated question answering, and curriculum content generation face potential risks of hallucinations that are especially harmful to users who lack sufficient background to recognize the errors of the model (Kaddour et al., 2023; Liang et al., 2022). Hallucination poses an especially big problem in the field due to the population asymmetry where the users do not have the required competence to critically evaluate the information presented by LLMs, which can lead to learning incorrect data by heart.

Frameworks used to detect misinformation (Beer and Matthee, 2020; Cavus et al., 2024; Thibault et al., 2025) have offered approaches that can be used in the context of educational moderation to detect explanations with factual errors regarding science and history. Beauvais (2022), Adams et al. (2023) have found that automation bias the inclination to take the outputs from an algorithm without questioning them is higher among users who have lower expertise within the specific domain.

In the context of education, the appropriate sources of evidence are already accepted curriculum standards, peer-reviewed textbooks, and reference material on the subject matter in question, as these provide the "ground truth" that must be reproduced by the tutoring/curriculum generation system (Beer and Matthee, 2020; Cavus et al., 2024). As the users of the tutoring system lack domain knowledge and therefore would be unlikely to detect errors made in the learning process, as well as the fact that incorrect information may become internalized by the learner, it is recommended to set a low level of tolerance for factual mistakes (dates, definitions, procedures) but higher tolerance for pedagogically open-ended statements.

5.5. Public information and journalism

The deployment of LLMs in news generation, content summarization, and public information dissemination creates macro-level hallucination risks that extend to societal trust in information ecosystems. Broda and Strömbäck (2024) have shown that the spread of misinformation undermines people's trust in journalism and democratic institutions. Tomassi et al. (2025) report how AI disinformation has been propagated within the discussion on climate science. Cavus et al. (2024) have developed a cloud-based system to detect fake news in online social networks in real-time.

6. Enabling technologies

Several technological enablers support the creation of hallucination verification systems. Similarly to the industrial technological enablers required to realize advanced production concepts (Lewis et al., 2020), verification technology could be considered the basic components from which multi-tier detection structures are created. The following is an overview of the key technological enablers discussed, presented in Table 5 and Fig. 3.

6.1. Transformer architectures and pre-trained language models

Building on the transformer architecture of Vaswani et al. (2017), the specific capability that transformer-based models contribute to hallucination verification is self-attention-based contextual representation of long-range dependencies, which is a necessary representational capability for claim decomposition, entailment analysis, and entity extraction in verification processes; the general architecture itself is not rederived here, as it is well established in the NLP literature. The importance of bidirectional pre-training for NLI fine-tuning was illustrated by BERT (Devlin et al., 2019); RoBERTa (Liu et al., 2019) improved model robustness; and DeBERTa (He et al., 2020) utilized disentangled attention, allowing for the best publicly available framework for entailment-based hallucination detection.

Larger models like GPT-3 (Brown et al., 2020), GPT-4 (Achiam et al., 2023), and LLaMA-2 (Touvron et al., 2023) play a double role in hallucination validation by providing the hallucinated text that needs to be validated and being used as the reference model for a multi-model consensus-based approach (Adlakha et al., 2024). Instruction tuning based on RLHF (Ouyang et al., 2022) and DPO (Rafailov et al., 2023) is an effective strategy in lowering the number of hallucinations by preference alignment. A detailed cross-family comparison of these architectural and alignment choices, and their implications for hallucination verification, is provided in Section 3.4 (Bhati et al., 2026).

6.2. Dense retrieval and retrieval-augmented generation

Retrieval-based techniques for hallucination mitigation using RAG embed queries and documents into common vector spaces based on semantic similarity for retrieval. These dense retrieval systems represent the core of current RAG-based techniques for mitigating hallucinations (Lewis et al., 2020; Asai et al., 2023). The main breakthrough achieved by the RAG architecture for addressing hallucinations is the exchange of parametric knowledge (which is prone to hallucinations) stored within the weights of the NLP model with non-parametric knowledge retrieved at inference time from external sources. This technique has been shown to effectively reduce the error rates in knowledge-intensive NLP tasks by Lewis et al. (2020). The Self-RAG technique (Asai et al., 2023) further improves upon this approach by enabling models

Table 5
Enabling technologies for hallucination verification systems and their roles across verification dimensions. Utilization levels: H = High, M = Medium, L = Low. Ratings follow the scoring protocol defined in Section 1 (paragraph preceding Table 1).

Technology	Factual Detect.	Faithful Detect.	Logical Detect.	Temporal Detect.	Entity Detect.	Key References
Transformer / Pre-trained LMs	H	H	H	M	H	Vaswani et al. (2017), Devlin et al. (2019), He et al. (2020)
Dense Retrieval / RAG	H	H	M	M	M	Lewis et al. (2020), Asai et al. (2023), Gao et al. (2023b)
Knowledge Graphs	M	L	H	H	H	Wang et al. (2020), Huang et al. (2025b), Shami et al. (2026)
Natural Language Inference	H	H	H	M	M	Zaman and Srivastava (2025), Devlin et al. (2019), He et al. (2020)
Credibility Scoring	H	M	L	L	M	Chrysidis et al. (2024), Akhtar et al. (2024), Srba et al. (2026)
Cross-Model Consensus	H	H	M	M	H	Shami et al. (2026), Adlakha et al. (2024), Manakul et al. (2023)
Self-Verification / Self-RAG	H	M	M	L	M	Asai et al. (2023), Manakul et al. (2023), Gao et al. (2023a)
Multimodal Fusion	M	M	L	L	M	Zhang et al. (2024b), Jiang et al. (2024), Wang et al. (2024)

to self-decide when retrieving information is necessary in order to reduce unnecessary retrieval overheads while retaining full coverage of the space. The RAGAS evaluation framework for RAG-based pipelines addresses their evaluation in terms of faithfulness, relevance, and recall (Es et al., 2024).

Based on recent research, dense retrieval using semantic vector embeddings, as provided by models such as sentence transformers, surpasses sparse retrieval methods like BM25 in tasks involving semantic similarity important for detecting hallucinations (Gao et al., 2023b; Xu et al., 2024). However, it has been shown by Mallen et al. (2023) that both parameterized and non-parameterized systems fail in the context of long-tail factual search. The distinction between such single-pass retrieval designs and iterative, tool-using agentic RAG pipelines is discussed in detail in Section 4.2.1.

6.3. Knowledge graphs and structured verification

Graphs of Knowledge – structured databases comprised of entities, attributes, and relations – allow the verification of claims by LLMs using symbolic verification based on grounded assertions. Wang et al. (2020) show how cross-graph representation learning can be used for knowledge graph validation, which constitutes the theoretical underpinning for the use of attribute-entity consistency verification in verifications pipelines. Huang et al. (2025b) propose an approach to graph-based verification that is specifically tailored to fact-checking, utilizing relational structure in order to propagate evidence through multi-hop claim chains. Shami et al. (2026) compare LLMs for knowledge graph verification, identifying a hybrid approach consisting of SPARQL-based querying and NLI entailments as the most accurate. Song et al. (2025) devise set-consistency energy networks, achieving competitive results in the context of structured consistency verification for multi-hop verification tasks.

The use of knowledge graphs within hallucination verification systems brings several strengths when compared to retrieval-based techniques, including computational efficiency for the verification of entity-attribute assertions, provision of traceable evidence pathways, and immunity to failure due to retrieval issues based on improper query formulation. Existing challenges include the insufficient coverage of specific domains within publicly available knowledge graphs like Wikidata and the cost of computational resources to maintain current entity representations within fast-changing domains.

6.4. Natural language inference and entailment scoring

The task of natural language inference (NLI), where it is needed to determine if a premise contradicts, entails, or is neutral with respect to a hypothesis, is an integral part of hallucination detection. Within the context of verification systems, the statement to be verified is the hypothesis and the gathered evidence is the premise. The entailment probability is an appropriate method of evaluating the consistency between the two and the degree of the factual relationship. DeBERTa (He et al., 2020), which was fine-tuned on MultiNLI dataset, provides cutting-edge performance when used for entailment scoring within the context of verification, reaching high accuracy when processing scientific and news domain texts. Zaman and Srivastava (2025) perform causal analysis of faithfulness metrics, finding factors which negatively influence the evaluation when only using NLI scores.

However, several papers (Pesaranghader and Li, 2026; Malin et al., 2025) find out that although the NLI-based entailment scoring is efficient in identifying factual hallucinations when there is unambiguous counter-evidence provided, it is rather unreliable when dealing with faithful and logical hallucinations with implicit contradiction or multi-hop reasoning.

6.5. Credibility scoring and source authority assessment

Evidence found in support of the claim and used for claim verification becomes reliable or unreliable based on the reliability of the evidence source. Srba et al. (2026) present an extensive review of automatic credibility assessment based on textual credibility cues, outlining the taxonomy of authority markers relevant to hallucination verification scenarios. Chrysidis et al. (2024) show that the ability to differentiate between credible and noncredible, or even leaked, pieces of evidence greatly boosts the accuracy of automated verdicts. Akhtar et al. (2024) propose the Ev2r measure as an evaluation measure for evidence retrieval-based fact-checking pipelines. Thus, the above-mentioned works prove that the authority of sources can be considered a valuable claim verification signal.

7. Comparative analysis of existing approaches

The following is an overview of comparative studies on existing hallucination detection and mitigation methodologies, with respect to their capabilities. In Table 6, the characteristics of some sample systems reported in the literature are presented on six different verification capabilities, while Fig. 4 shows the multi-layer verification architecture adopted by contemporary methodologies. Similar to the H/M/L rating used in Section 1 for Table 1, the Yes/Partial/No rating in Table 6 uses a clear criterion as opposed to an implicit one: The capability gets a rating of *Yes* when it is a core component of the system being described and evaluated; *Partial* if the capability is there in an optional way that may or may not be well-evaluated (e.g., supported but not explicitly tested, or tested only as a side-effect of some other contribution); and *No* if the capability does not exist in the cited work at all. Similar to the H/M/L rating protocol, this Yes/Partial/No rating was assigned manually, through a careful reading of the papers in question, and was not done using any multi-rater independent procedure, as was noted in Section 9.3.

7.1. Single-layer versus multi-layer verification

The most basic division among approaches described in the literature can be made between single-layer solutions using only one type of evaluation method such as RAG retrieval or NLI evaluation and multi-layer approaches combining several independent evaluation criteria.

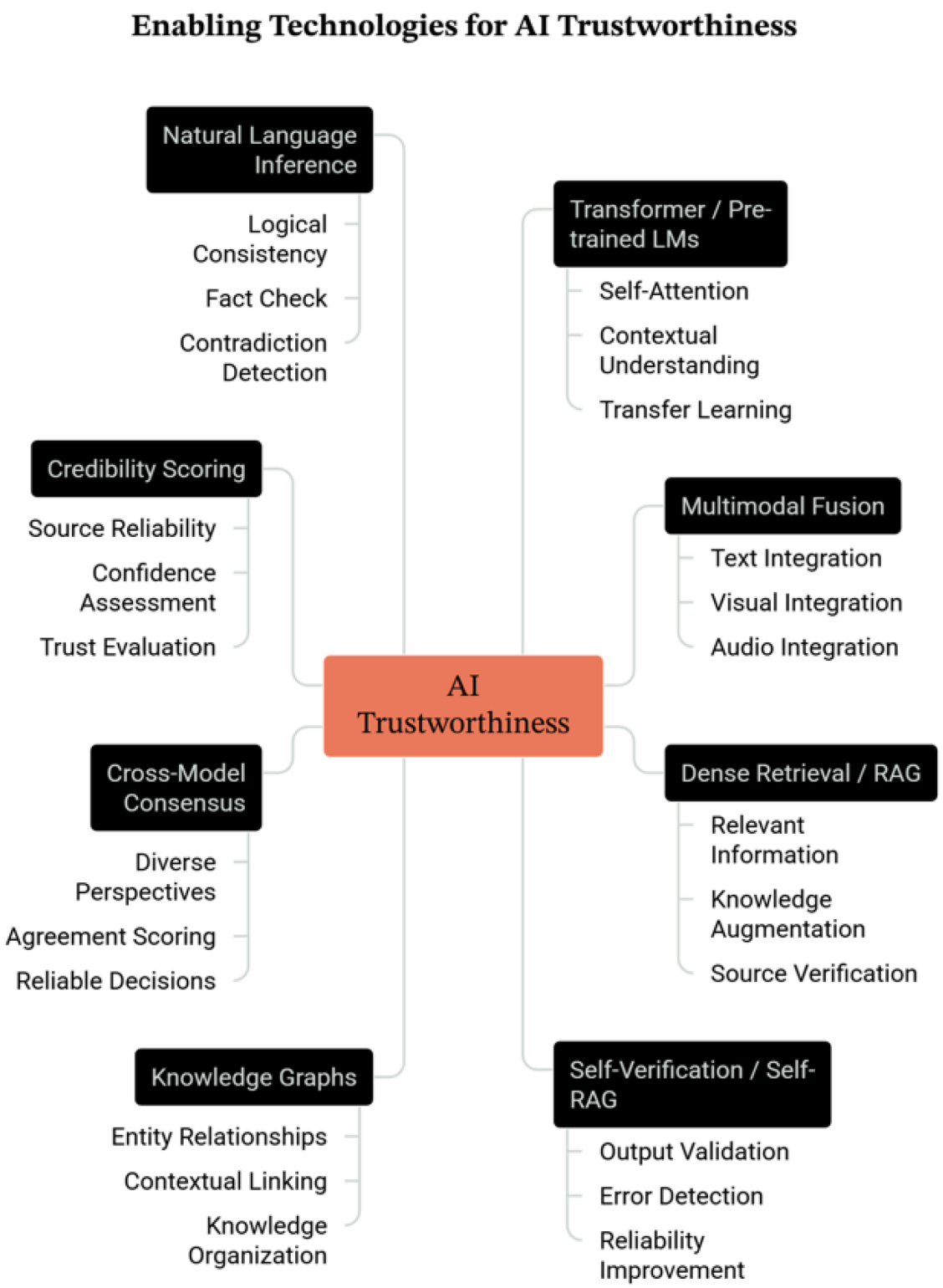


Fig. 3. Key enabling technologies of hallucination verification systems and their inter-relationships within multi-layer verification architectures.

Single-layer solutions such as FActScore (Min et al., 2023b) or QAFactEval (Fabbri et al., 2022) provide efficiency and interpretability but are inherently limited to detecting only the specific hallucination types for which they were designed. Ramprasad and Wallace (2024) show how single-metric approaches systematically overestimate accuracy for domain evaluation tasks, thus giving practical reason to adopt multi-layer approaches.

Empirical evidence consistently demonstrates that an ensemble of retrieval, cross-model comparison, and logical consistency signals provides superior performance compared to any single-layer approach (Manakul et al., 2023; Gao et al., 2023a). Furthermore, ablation studies (Manakul et al., 2023; Gao et al., 2023a) demonstrate that the removal of both retrieval and cross-model comparison components results in substantial degradation of verification performance, confirming

Table 6
Comparative analysis of representative hallucination detection and verification systems across key capability dimensions. Yes/Partial/No rating criteria are defined in Section 7, paragraph preceding this table.

System / Method	Year	Halluci. Detect.	RAG / Knowledge	Cross-Model	Truth Score	Auto-Correct	Multilingual	Key Reference
TruthfulQA	2022	Yes	No	No	Partial	No	No	Lin et al. (2022)
FActScore	2023	Yes	Partial	No	Yes	No	No	Min et al. (2023b)
QAFactEval	2022	Yes	Partial	No	Yes	No	No	Fabbri et al. (2022)
FEQA	2020	Yes	No	No	Partial	No	No	Durmus et al. (2020)
Self-RAG	2023	Partial	Yes	No	No	Partial	No	Asai et al. (2023)
RAGAS	2024	Partial	Yes	No	Partial	No	No	Es et al. (2024)
RAGTruth	2024	Yes	Yes	No	Partial	No	No	Niu et al. (2024)
AVerITeC	2023	Yes	Yes	No	No	No	No	Schlichtkrull et al. (2023)
ClaimCheck	2025	Yes	Yes	No	Partial	No	No	Putta et al. (2025)
SelfCheckGPT	2023	Yes	No	Partial	No	No	No	Manakul et al. (2023)
RARR	2023	Yes	Yes	No	No	Yes	No	Gao et al. (2023a)
Multi-Agent FC	2026	Yes	Yes	No	No	No	No	Gong et al. (2026)
MultiClaimNet	2025	Yes	No	No	No	No	Yes	Panchendrarajan et al. (2025)
KG Verification	2025	Partial	No	No	Partial	No	No	Huang et al. (2025b)

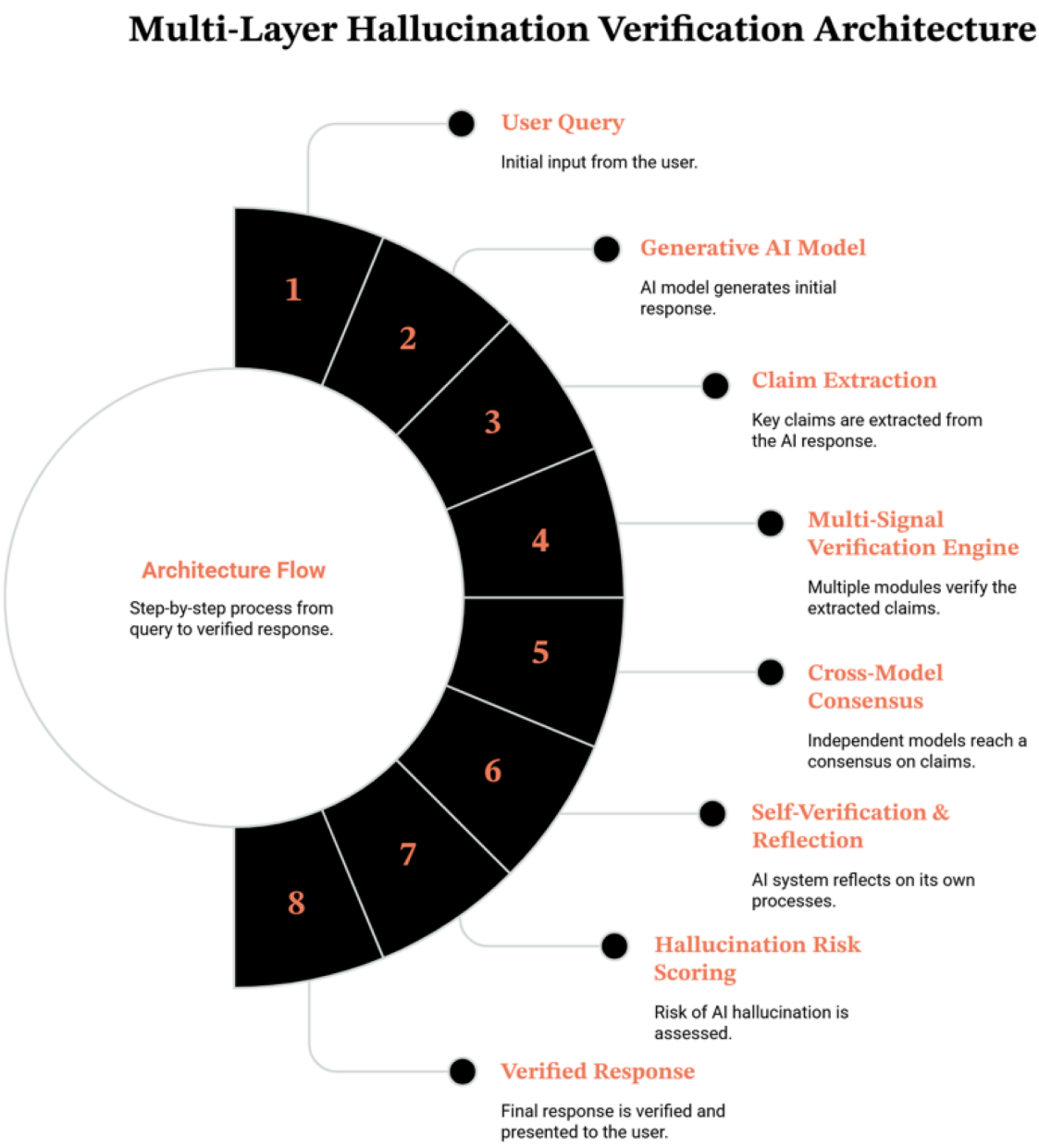


Fig. 4. Conceptual multi-layer hallucination verification architecture synthesized from the surveyed literature, illustrating the convergent multi-signal design pattern that characterizes state-of-the-art verification systems.

their complementary contributions to robust detection.

The multi-layer approach shown in Fig. 4 relies on a combination of evidence gathered from retrieval, cross-model comparison, NLI entailment, and knowledge-graph validation, but the literature synthesized here does not suggest any single strategy for handling the conflicts between these evidence types - such as when the evidence from retrieval suggests a certain claim is true while an NLI model finds a contradiction, or when there is a consensus across models that disagrees with a knowledge-graph evaluation. Three conflict resolution strategies can be inferred from the layer-specific approaches discussed elsewhere in this survey despite the fact that none of the surveyed systems employ all three: (i) *precedence rules* - a form of hard-wired policy that makes a certain type of signal more authoritative over other types when resolving conflicts for a specific type of hallucination, e.g., precedence of knowledge-graph validation for entity-attribute claims (Section 6.3) due to its higher rating for the type of hallucination (Table 5), and precedence of NLI entailment for faithfulness claims; (ii) *uncertainty aggregation*, i.e., making every layer output a calibrated measure of uncertainty rather than a binary classification, and combining these uncertainties using either a weighted aggregation (which could be optimized in a data-driven fashion as suggested in Section 9.2), or a Bayesian one; and (iii) *human review*, i.e., escalation of conflict resolution to a human expert in case the degree of disagreement between the signals passes a certain threshold. In the absence of a validated conflict resolution algorithm in the current literature, the latter is the most conservative and currently best-supported default approach in high-stakes settings (Section 5), while the two former are more applicable in low-stakes, high-throughput situations.

7.2. Retrieval-grounded versus model-internal approaches

Another important dichotomy is between retrieval-grounded verification, which grounds its claims on externally retrieved information, and verification methods that utilize consistency signals generated by the model itself. RAG models (Lewis et al., 2020; Asai et al., 2023; Gao et al., 2023b) and fact-checking pipelines (Guo et al., 2022; Schlichtkrull et al., 2023; Putta et al., 2025) fall under the former category while self-consistency-based methods (Manakul et al., 2023) and model self-evaluation (Pesaranghader and Li, 2026) belong to the latter one.

While verification methods based on externally retrieved information benefit from independence from biased knowledge representation of the generating model, their performance is limited by retrieval quality and coverage (Mallen et al., 2023; Niu et al., 2024). Model internal verification approaches, on the other hand, can identify hallucinations independently of external APIs but suffer from correlated errors due to consistent model misrepresentations in certain domains (Bommasani et al., 2021; Shami et al., 2026). Current literature points to the benefits of hybrid approaches that prefer retrieval-based signals to cross-model consensus in detecting diverse hallucinations.

The model-agnostic framing adopted throughout this survey (Section 1) refers specifically to *parameter* access: post-generation middleware of this kind does not require access to the generating model's weights, training data, or internal activations. This should not be read as implying that such middleware is agnostic to the generating model in every respect. In practice, the effectiveness of externally applied verification still depends on properties of the generating model's *output*: its verbosity and formatting conventions affect how cleanly a response can be decomposed into independently verifiable claims (Min et al., 2023b); its citation behavior (whether and how it attributes claims to sources) determines how much verification work the middleware must perform versus can offload to the model itself; and its output language determines which retrieval, NLI, and knowledge-graph resources are available downstream, a constraint discussed further in Section 8 on multilingual coverage. Consequently, "model-agnostic" is more accurately understood as *parameter-agnostic but not output-agnostic*: the middleware can be applied to any generating model without modification to that model, but its practical performance is still shaped by that model's output characteristics, and this distinction should be kept in mind when comparing verification systems across different generating models.

7.3. Factuality metrics: automated versus human evaluation

The difference between automated metrics of factuality and human judgment is an enduring problem in the literature. As Ramprasad and Wallace (2024) show, automated metrics tend to overestimate the accuracy of large language models, while Zhang et al. (2024a) report contradictions between automated measures of faithfulness and human citation judgment. According to the literature reviews by Lamsiyah et al. (2025) and Malin et al. (2025), there is no automated metric that covers all aspects of factual accuracy in different application areas and tasks.

Such evidence inspires the idea of combining multiple automated metrics with human assessment on sample datasets (Jacovi et al., 2025). The results of causality analysis conducted by Zaman and Srivastava (2025) indicate that some factors affecting the evaluation of a single metric may lead to the wrong estimate of the model reliability. HELM (Liang et al., 2022) is currently the most ambitious framework for evaluating different metrics such as factuality, toxicity, fairness, and efficiency at once.

7.4. Domain specificity and generalization

Different solutions exhibit wide variations in terms of domain. General-purpose verification solutions like TruthfulQA (Lin et al., 2022), AVerITeC (Schlichtkrull et al., 2023), and RAGAS (Es et al., 2024), for example, offer extensive benchmarking capabilities but might fail to accommodate the specialized nature of knowledge required by verification in medical, legal or scientific settings. On the other hand, domain-specific approaches such as solutions designed to detect fake news within particular regions (Kishwar and Zafar, 2023) and to verify multimodal content in e-commerce (Liu et al., 2024) exhibit greater precision but lack generalizability.

The recent body of work shows that the choice of the retrieval sources has a strong impact on the ability of a RAG solution to handle various domains: systems with access to specialized academic resources and institution-specific repositories outperform other systems at detecting domain-specific hallucinations significantly (Chrysidis et al., 2024; Srba et al., 2026). These findings suggest that scalable domain-specific hallucination detection requires customized retrieval sources tailored to specialized knowledge repositories, rather than improvements to base model architectures alone.

8. Open challenges and research gaps

Despite advancements in detecting and mitigating hallucinations, there are certain challenges within the research literature that constitute the current frontier of research within the field. The following is a synthesis of these key open challenges and research gaps.

8.1. Scalability and computational efficiency

The multi-layer verification architectures that achieve the best detection performance are computationally expensive, typically requiring multiple API calls to large generative models, dense vector retrieval across large corpora, and knowledge graph lookups (Gao et al., 2023b; Shami et al., 2026). The end-to-end delay latency of existing frameworks often surpasses five seconds, making it impractical for real-time generation processes as well as browser-based applications (Asai et al., 2023; Manakul et al., 2023). According to the current literature, evidence caching with time-to-live (TTL) expiration policies for commonly verified themes, asynchronous processing of verification stages, and fast mode for certain claims can be considered promising techniques for lowering latency costs (Es et al., 2024; Xu et al., 2024).

8.2. Temporal inconsistency and knowledge staleness

Temporal hallucinations, which involve LLMs stating facts that are valid at training time but subsequently become obsolete, constitute an especially tricky type of hallucination that is poorly handled by most current solutions (Pesaranghader and Li, 2026; Drchal et al., 2024). Temporal staleness can be countered by RAGs via retrieval of current evidence; however, the success of such an approach hinges on the freshness and scope of the retrieval corpus (Gupta et al., 2024; Gao et al., 2023b). Knowledge graphs are also subject to temporal staleness when attributes of entities are updated faster than knowledge graph updates (Wang et al., 2020; Shami et al., 2026). The current state-of-the-art lacks a robust method of detecting temporal hallucinations applicable to fast-evolving fields like politics, finance, and medicine.

There are two failure modes that both take the form of an LLM asserting a now out-of-date fact, which require different types of mitigation strategies. Proper temporal hallucination, as outlined in Table 2, is where an LLM asserts an incorrect date, duration, or chronological order, often arising from a training-time snapshot that is not kept up-to-date. A second, distinct type of failure is *the retrieval of outdated but formerly correct evidence*, where a RAG or verification process finds a passage that was accurate at its time of publication, but which has since been rendered obsolete by later events (for example, an outdated clinical guideline, a repealed statute, or a superseded office-holder record), and the verification system mistakenly marks the information as still being current. The first is a property of the generating LLM's knowledge base; the second is a property of the retrieved evidence itself and doesn't require any LLM hallucinations at all. Since the second failure mode is not addressed by conditioning the generation process on the retrieved facts alone, there are three specific types of mitigations found in the wider literature on RAG and knowledge graphs that are directly applicable: *timestamped retrieval*, where evidence is indexed and ranked by recency and where older-than-threshold passages are ignored or down-weighed; *temporal knowledge graphs*, where entity attributes are stored with versioning information, allowing a verification system to distinguish between facts that were true as of some past t, and those that are true now (Wang et al., 2020; Shami et al., 2026); and *evidence expiry and re-verification policies*, where claims are assigned a validity period after which they are expired and must be re-verified rather than being treated as permanently verified. No systems found in Table 6 use all three forms of mitigation together, which is highlighted by this survey as an important area for future research and benchmarking.

8.3. Interpretability and explainability

Present systems for detecting hallucinations generate verdicts about their validity that remain inaccessible to the lay user; they provide an aggregated measure of reliability without specifying the particular claims that were flagged and the reason for flagging (Jeong et al., 2025; Thorne, 2024). According to Thorne, interpretability is a pre-requisite for correct calibration of user trust regarding verification output as it makes no sense to act responsibly when dealing with results whose meaning cannot be understood (Thorne, 2024). Jeong et al. show that transparency and interpretability constitute basic conditions for a reliable AI system (Jeong et al., 2025). However, the majority of systems designed for hallucination detection still provide only aggregated measurements.

8.4. Domain dependency and specialization

The performance of both retrieval-grounded and model-based verification approaches degrades substantially in specialized domains where training data coverage is limited and specialized knowledge is required for accurate verification. Similarly, Niu et al. (2024) argue that RAG models are likely to hallucinate when they rely on insufficiently specialized information. Medical, legal, and financial fields have their specific terminology, referencing style, and standard requirements for evidence, which are not taken into account by generalized verification tools (Kaddour et al., 2023; Bommasani et al., 2021). Comprehensive domain-adaptive verification frameworks that systematically address these specialized requirements remain an understudied area, representing a critical gap for future research.

8.5. Multilingual coverage

The vast amount of research regarding hallucination detection has been carried out in English with multilingual approaches receiving much less focus (Panchendrarajan and Zubiaga, 2026; Panchendrarajan et al., 2025; Drchal et al., 2024). An example of work that stands out in this regard is the work by Panchendrarajan and Zubiaga (2026), who present cross linguistic claim detection that is aware of entities at the EACL 2026 conference. The authors Panchendrarajan et al. (2025) present MultiClaimNet as a multilingual tool to evaluate clusters of fact checked claims. The authors (Drchal et al., 2024) propose an approach to automated fact checking in multiple languages.

8.6. Multimodal hallucination

As LLMs increasingly incorporate visual, auditory, and structured data inputs, hallucination extends beyond text to encompass multimodal content. Jiang et al. (2024) propose a hallucination enhanced contrastive learning technique for multimodal LLMs in CVPR 2024, illustrating that multimodal hallucinations happen in different ways compared to hallucinations in text-based LLMs. Zhang et al. (2024b) evaluate factuality on multimodal summarization, stressing the problems encountered in verifying multimodal hallucinations outside text. Wang et al. (2024) introduce multimodal fake news detection through LLMs. Nevertheless, multimodal benchmark datasets that tackle all three types of problems together require further development.

8.7. Privacy-preserving verification

LLM hallucination verification systems typically require transmitting user queries and AI-generated responses to external modules, including retrieval APIs, cross-model evaluation APIs, and knowledge graph services (Lewis et al., 2020; Gao et al., 2023b; Shami et al., 2026). This process raises significant privacy concerns in critical domains such as healthcare and legal practice, where user inputs may contain personally identifiable information. According to Weidinger et al. (2021), privacy becomes an ethical issue with respect to LLMs, while Bommasani et al. (2021) observe that while verification modules are necessary for the proper functioning of LLMs, they complicate data governance considerably. In the current literature, there is no architectural design for verifying LLM outputs without transmitting sensitive data externally (refer to Table 7).

8.8. Human-in-the-loop verification

Most of the verification architectures discussed in Section 7 are fully automated end-to-end, despite the domain-specific error-threshold analysis presented in Section 5 implying that fully automated decisions might be unsuitable for high-risk and truly ambiguous claims (for example, disputed legal precedent, emerging clinical recommendation, or a claim contradicting evidence presented in Section 7). However, the surveyed literature gives little guidance on human review integration into the verification cycle. Automation bias – the phenomenon of the user, especially the non-expert user, trusting algorithmic output blindly (Beauvais, 2022; Adams et al., 2023) – is the key challenge here: adding a human reviewer to the process without showing him or her the evidence underlying the verdict does not mitigate the bias, as the literature on claim-level evidence presentation and interpretability (Jeong et al., 2025; Thorne, 2024) emphasizes. As we adopt the concept of interpretability from Thorne (Thorne, 2024) in Section 3.5, the human review needs to have access not only to the aggregated confidence score

Table 7
Open research challenges and potential future directions in hallucination detection and mitigation.

Challenge	Current Limitation	Promising Future Direction
Scalability and Latency	Multi-layer pipelines exceed 5 s per query (Gao et al., 2023b; Shami et al., 2026)	Evidence caching; parallel async execution; early-exit fast mode (Asai et al., 2023; Es et al., 2024)
Temporal Inconsistency	Knowledge staleness in RAG and KG systems (Gupta et al., 2024; Wang et al., 2020)	Real-time retrieval indexing; temporal hallucination-specific benchmarks (Drchal et al., 2024)
Interpretability	Aggregate scores without claim-level evidence (Jeong et al., 2025; Thorne, 2024)	Claim-highlighting interfaces; evidence attribution dashboards (Jeong et al., 2025; Thorne, 2024)
Domain Dependency	Performance degradation in medical / legal domains (Mallen et al., 2023; Niu et al., 2024)	Domain-tailored retrieval portfolios; domain-specific NLI fine-tuning (Chrysidis et al., 2024)
Multilingual Coverage	English-language bias across benchmarks (Panchendrarajan and Zubiaga, 2026; Drchal et al., 2024)	Cross-lingual NER/NLI; low-resource multilingual benchmarks (Panchendrarajan et al., 2025)
Multimodal Extension	Text-only verification inadequate for multimodal LLMs (Zhang et al., 2024b; Jiang et al., 2024)	Visual grounding; audio verification; structured data consistency (Wang et al., 2024; Abduljaleel and Ali, 2024)
Privacy Preservation	External API calls transmit sensitive query content (Bommasani et al., 2021; Weidinger et al., 2021)	Federated verification; on-device NLI models; local knowledge graphs (Hendrycks et al., 2020)
Weight Calibration	Expert-defined composite score weights lack empirical grounding (Ramprasad and Wallace, 2024; Malin et al., 2025)	Bayesian regression weight optimization over annotated datasets (Muhlgay et al., 2024)
Human-in-the-Loop Verification	No systematic guidance on when human adjudication is needed, what evidence to show reviewers, or how to reduce automation bias (Beauvais, 2022; Adams et al., 2023)	Evidence-grounded review interfaces; disagreement-driven escalation policies (Section 8.8)
Coverage–Usability Trade-off	No explicit policy for varying verification depth by context in surveyed systems (Table 6)	Risk-adaptive verification depth by domain, claim importance, and consequence of error (Section 8.9)

but to the evidence underlying the verdict. Identifying when the human intervention is necessary (e.g., via conflicting signals criteria discussed in Section 7), reducing the reviewer's load without automating him out of the process, and resolving the disagreements between the expert reviewer and the automated verdicts are additional open challenges, not covered in the surveyed literature, and, thus, listed among the research priorities alongside with the governance issues mentioned in Section 9.2.

8.9. Coverage-usability trade-offs and risk-adaptive verification

An open challenge related to, but different from the latency issue discussed in Section 8.1 is the trade-off between the verification *coverage* and *usability*. Independent verification of all the atomic claims composing the response, as in the case of the FActScore-style atomic decomposition (Min et al., 2023b), increases the factual coverage of the response, but implementing such approach to all the claims in all the responses can fragment the response, generate excessive warnings to the user, degrade user experience, and increase latency and computational costs, as described in Section 8.1. None of the compared systems listed in Table 6 explicitly describe any policy of verification depth adaptation. Given the domain-specific error thresholds established in Section 5, the solution appears obvious: *risk-adaptive verification*, in which the depth of the claim verification (e.g., full atomic decomposition or single holistic verification) is determined based on the domain (medical and legal claims deserve deeper verification than casual information search), importance of the claim to the response (the more important the claim is, the deeper the verification needs to be performed), the uncertainty of the model (confidence score), the expertise of the user (and thus the amount of risk that can be left on the user), and consequences of the false answer being uncaught. Implementation of the risk-adaptive verification in practice depends on solving the human-in-the-loop problem stated in Section 8.8 and conflicting signals handling strategy of Section 7.

9. Lessons learned and future research directions

This section discusses the lessons learned from the current state-of-the-art studies, and based on these lessons, synthesizes the future research challenges that need to be addressed to pave the way for reliable, interpretable, and scalable hallucination detection and mitigation ecosystems in next-generation generative AI systems.

9.1. Lessons learned

Rapid progress in generative AI has brought forward the increasing relevance of mechanisms aimed at hallucination detection, validation, and mitigation in various spheres of application such as medicine, education, science, law, and informational services for the general public. It is clear from the findings of the reviewed literature that there is no universal method for detecting and validating hallucinations, and therefore it is crucial to develop a multi-tier system of mechanisms that include data retrieval, fact checking, reasoning and trustworthiness assessment methods. This section outlines key lessons learned from cutting-edge research, technologies, and real-life implementation cases of generative artificial intelligence.

9.1.1. On hallucination taxonomy

Based on the definitions and categorizations provided in the literature,it can be seen that hallucinations are multidimensional concepts that cannot be defined by one type of error only. According to Huang et al. (2025a), factuality, faithfulness, and consistency are three different failures that have different origins and requirements. In turn, Pesaranghader and Li (2026) have expanded the taxonomy into four types of errors, including factual, faithful, logical, and temporal. The common theme in all these studies is that the design of a detection algorithm for one specific error type would not work for other error types.

9.1.2. On detection methodologies

The surveyed literature consistently shows that there exists no single detection method that performs reliably across all types of hallucination and across all application domains. RAG systems decrease but do not eliminate hallucinations (Niu et al., 2024); factuality metrics overestimate accuracy systematically (Ramprasad and Wallace, 2024); cross-model agreement is constrained by the extent of overlap in training data (Bommasani et al., 2021); and self-evaluating models are subject to self-referential bias (Pesaranghader and Li, 2026). One of the strongest lessons from the literature is that verification systems that combine retrieval, cross-model agreement, and logical coherence layers significantly outperform single-layer systems, a result that has been confirmed repeatedly in ablation studies (Manakul et al., 2023; Gao et al., 2023a).

9.1.3. On enabling technologies

Enabling technologies such as transformer-based NLI, dense retrieval, knowledge graphs, and credibility scoring each contribute distinct verification abilities. Transformer NLI systems produce claim-level

entailment scores (Devlin et al., 2019; He et al., 2020); dense retrievers ensure that verification is grounded in current external knowledge (Lewis et al., 2020; Asai et al., 2023); knowledge graphs enable entity-attribute consistency checks (Wang et al., 2020; Huang et al., 2025b; Shami et al., 2026); and credibility scorers weight evidence based on source authority (Chrysidis et al., 2024; Srba et al., 2026). The use of enabling technologies in combination is what defines the state-of-the-art approach in hallucination verification system design.

9.1.4. On application domains

Analysis of verification across various application domains suggests that false positive rates, hallucination types, and acceptable rates of corrections differ significantly between them. In healthcare and legal domains, very low false positives rates are needed to correct hallucinations, while information access applications can accept higher false positive rates in return for higher recall (Kaddour et al., 2023; Bommasani et al., 2021; Weidinger et al., 2021). This need to account for the variation in performance criteria motivates the development of configurable verification systems that allow for adjusting verification performance based on context.

9.2. Future research directions

Although there has been significant progress towards detecting and mitigating hallucinations in AI-generated content, there are some challenges left to be solved, such as scalability, computational efficiency, explainability, multilingual capability, privacy preservation, and multimodel verification. From reviewing the relevant studies, it is possible to conclude that existing methods have their limitations in terms of providing consistent, explainable, and domain-adaptive verification mechanisms. Taking into account identified gaps and innovative developments, this part will address potential areas of future research for the development of more efficient verification methods.

9.2.1. Data-driven weight optimization

A limitation that is common across composite trust scoring approaches is that weight assignments are typically made based on expert judgment rather than data-driven optimization (Ramprasad and Wallace, 2024; Malin et al., 2025). A future research direction would therefore be to develop methodologies to optimize component weights using, for example, Bayesian regression techniques on verification datasets (Muhlgay et al., 2024). Developing weight vectors for medicine, law, and finance would be of particular importance given the performance degradation seen in these domains.

9.2.2. Multilingual and cross-Lingual verification

It is necessary to overcome the heavy English-language bias of existing research on hallucination detection through the multilingual expansion of benchmarks and verification systems. Cross-lingual entity-aware claim detection (Panchendrarajan and Zubiaga, 2026) and multilingual claim clustering (Panchendrarajan et al., 2025) lay important foundations, but multilingual components for NER, NLI, and retrieval are necessary for verification systems to operate across multiple languages. Multilingual fact checking infrastructure such as MultiClaimNet (Panchendrarajan et al., 2025) and multilingual fact checking pipelines (Drchal et al., 2024) should be extended to include lower-resource languages absent from the literature.

9.2.3. Real-time verification and latency reduction

Verification systems will have to reduce end-to-end verification latency considerably before practical deployment is possible. A promising avenue of future research would be the use of pre-computed embeddings for topics that are frequently queried, progressive user interfaces that provide results before full verification has taken place, local device-based NLI systems for privacy-sensitive use cases, and adaptive verification depth that allocates computational resources proportional to query complexity and domain-specific risks (Asai et al., 2023; Manakul et al., 2023; Gao et al., 2023a).

9.2.4. Open-source and privacy-preserving architectures

Proprietary LLM APIs limit the cost, privacy, and reproducibility of verification systems considerably. Future research could systematically analyze the potential of open-source models such as LLaMA 2 (Touvron et al., 2023) as drop-in replacements in cross-model verification systems, evaluating the performance-cost trade-off in these systems. Federated verification approaches that localize entity-grounding and entailment scoring while preserving the confidentiality of input queries is an important avenue for verification within privacy-sensitive contexts.

9.2.5. Multimodal hallucination verification

The extension of verification systems to multimodal content, including images, audio, and structured data, is an inevitable necessity for the growing capabilities of generative models. Augmented hallucination learning for multimodal LLMs (Jiang et al., 2024) and multimodal factuality evaluation (Zhang et al., 2024b) establish important foundations, but benchmark datasets and end-to-end verification pipelines for multimodal claims remain severely underdeveloped in the existing literature.

9.2.6. Alignment, ethics, and governance

Future research should focus on addressing the ethical implications of verifying the output of LLMs, including automation bias (Beauvais, 2022; Adams et al., 2023), governance of verification decisions in high-stakes contexts (Jeong et al., 2025; Thorne, 2024), and aligning verification systems with diverse cultures and epistemic norms (Hendrycks et al., 2020; Perez et al., 2022). The research by Perez et al. (2022) shows that even aligned models can generate hallucinations under adversarial prompting conditions, establishing red teaming of verification systems as an important part of the verification process. The issue of verification infrastructure governance which includes questions such as control over evidence sources and credibility tier assignment and correction decisions, is a neglected sociotechnical problem that requires interdisciplinary collaboration.

9.3. Directions for strengthening this survey's methodology

In addition to the open questions for future research inherent in the surveyed literature itself (Section 8), the review methodology on which this survey is based (Section 2) presents a number of avenues for improving the robustness of any future iterations of this analysis. To start with, future updates to this survey could conduct a pre-registered systematic review against all included literature to address the *publication bias* whereby studies presenting novel or positive findings tend to be more readily published and indexed than studies presenting null or negative findings. Secondly, broadening the search protocol to venues outside the English language could help future studies take into account any verification research not covered by the English-language literature considered in this survey. Next, given the rapid developments occurring in the field of hallucination detection and verification, one promising approach going forward is to create a living version of this survey by periodically updating the search protocol cut-off date (Section 2). Additionally, future replication of the comparative analysis conducted in this survey (Section 1.2) could use an independently validated multi-rater score as well as the structured author judgments conducted in this study. Finally, as discussed in Section 9.2, conducting the full empirical evaluation of the TruthGuard AI architecture proposed in this survey (Section 1) is a critical next step towards validating this concept.

10. Conclusion

This survey has presented a comprehensive review of hallucination detection, evaluation, mitigation, and verification research in generative AI systems, synthesizing literature spanning 2016 through 2026

across eight major research dimensions. We established definitional clarity around hallucination types, healthcare, law, scientific research, education, and public information. We then reviewed state-of-the-art methodologies across retrieval-augmented generation, automated fact-checking, natural language inference, language modeling, misinformation detection, cross-model verification, and trustworthy AI frameworks. Key application domains, including healthcare, law, scientific research, education, and public information, were examined alongside their enabling technologies, such as dense retrieval, knowledge graphs, and credibility scoring. A comparative analysis revealed the superiority of convergent multi-signal architectures over single-layer approaches, a persistent gap between automated metrics and human evaluation, and substantial domain dependence. Several consistent lessons emerge. Hallucination is a multi-dimensional phenomenon resisting single verification methodologies. Combining retrieval-based, cross-model, and logical consistency signals provides the most robust detection. Enabling technologies are complementary rather than competitive, with each addressing distinct hallucination categories that other technologies cannot cover alone. Furthermore, application domains determine necessary precision–recall trade-offs, demanding configurable architectures. Finally, open challenges like scalability, temporal inconsistency, interpretability, and multilingual coverage define a critical research agenda. Systematic hallucination verification remains a societal imperative, and the research agenda outlined in this survey provides a structured pathway toward trustworthy, interpretable, and domain-adaptive generative AI systems.

CRediT authorship contribution statement

Syed Muhammad Waqas: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Ziyan Umer:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Affan Alim:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Muhammad Arham:** Writing – review & editing, Writing – original draft, Resources, Methodology, Formal analysis, Data curation, Conceptualization; **Mazliham Mohd Su'ud:** Writing – review & editing, Software, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Data curation; **Syed Jafar Ali Kazmi:** Writing – review & editing, Software, Resources, Investigation, Formal analysis, Data curation, Conceptualization; **Muhammad Mansoor Alam:** Writing – review & editing, Software, Investigation, Formal analysis, Data curation, Conceptualization.

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Data availability

Data will be made available on request.

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